

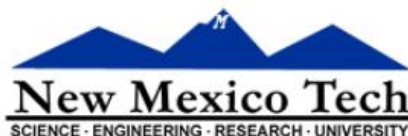
Imaging and Deconvolution

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Fourteenth Synthesis Imaging Workshop

2014 May 13– 20



References

- Thompson, A.R., Moran, J.M., Swensen, G.W. 2004
“Interferometry and Synthesis in Radio Astronomy”, 2nd edition (Wiley-VCH)
- previous Synthesis Imaging Workshop proceedings
 - Perley, R.A., Schwab, F.R., Bridle, A.H. eds. 1989 ASP Conf. Series 6
“Synthesis Imaging in Radio Astronomy” (San Francisco: ASP)
 - Ch. 6 Imaging (Sramek & Schwab) and Ch. 8 Deconvolution (Cornwell)
 - www.aoc.nrao.edu/events/synthesis
 - lectures by Cornwell 2002 and Bhatnagar 2004, 2006
- IRAM Interferometry School proceedings
 - www.iram.fr/IRAMFR/IS/IS2008/archive.html
 - Ch. 13 Imaging Principles and Ch. 16 Imaging in Practice (Guilloteau)
 - lectures by Pety 2004-2012
- many other lectures and pedagogical presentations are available
 - ALMA primer, ATNF, CARMA, ASIAA, e-MERLIN, ...



Visibility and Sky Brightness

- $V(u,v)$, the complex visibility function, is the 2D Fourier transform of $T(l,m)$, the sky brightness distribution (for incoherent source, small field of view, far field, etc.)
[for derivation from van Cittert-Zernike theorem, see TMS Ch. 14]

- mathematically

$$V(u, v) = \int \int T(l, m) e^{-i2\pi(ul+vm)} dl dm$$

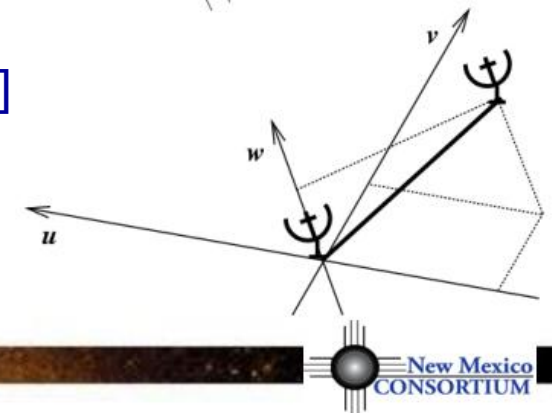
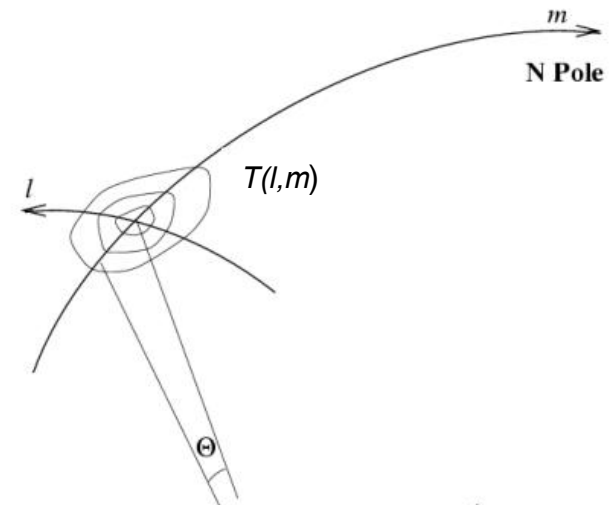
$$T(l, m) = \int \int V(u, v) e^{i2\pi(ul+vm)} du dv$$

u, v are E-W, N-S spatial frequencies [wavelengths]

l, m are E-W, N-S angles in the tangent plane [radians]

(recall $e^{ix} = \cos x + i \sin x$)

$$V(u, v) \xrightarrow{\mathcal{F}} T(l, m)$$

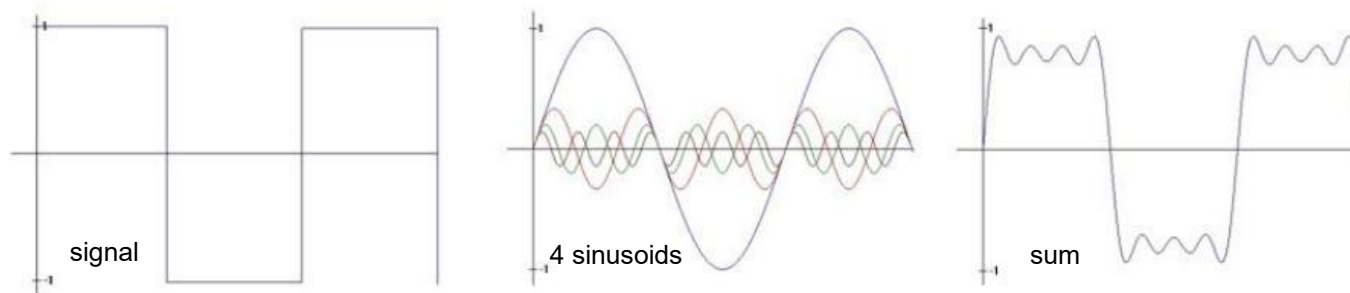


The Fourier Transform

- Fourier theory states and any well behaved signal (including images) can be expressed as the sum of sinusoids



**Jean Baptiste
Joseph Fourier**
1768-1830



$$x(t) = \frac{4}{\pi} \left(\sin(2\pi ft) + \frac{1}{3} \sin(6\pi ft) + \frac{1}{5} \sin(10\pi ft) + \dots \right)$$

- the Fourier transform is the mathematical tool that decomposes a signal into its sinusoidal components
- the Fourier transform contains *all* of the information of the original signal



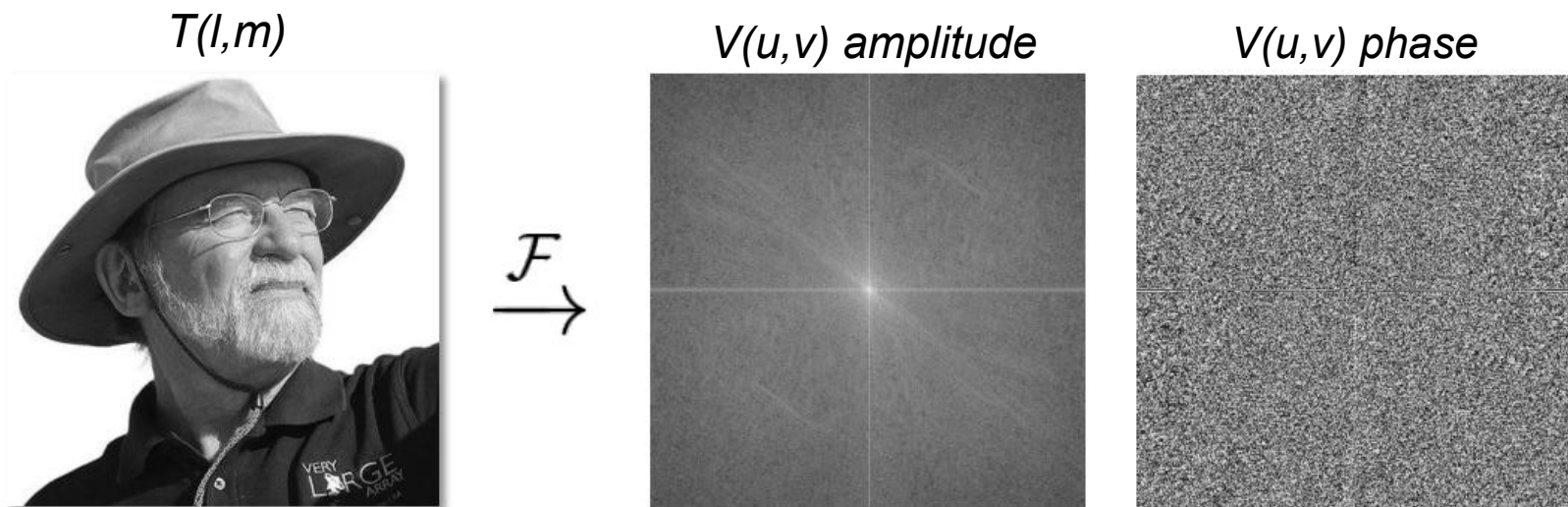
The Fourier Domain

- acquire some comfort with the Fourier domain
 - in older texts, functions and their Fourier transforms occupy *upper* and *lower* domains, as if “functions circulated at ground level and their transforms in the underworld” (Bracewell 1965)
 - some properties of the Fourier transform $g(x) \xrightarrow{\mathcal{F}} G(s)$
 - adding $g(x) + h(x) = G(s) + H(s)$
 - scaling $g(\alpha x) = \alpha^{-1} G(s/\alpha)$
 - shifting $g(x - x_0) = G(s) e^{i2\pi x_0 s}$
 - convolution/multiplication $g(x) = h(x) * k(x) \quad G(s) = H(s)K(s)$
 - Nyquist-Shannon sampling theorem
- $g(x) \subset \Theta$ completely determined if $G(s)$ sampled at $\leq 1/\Theta$

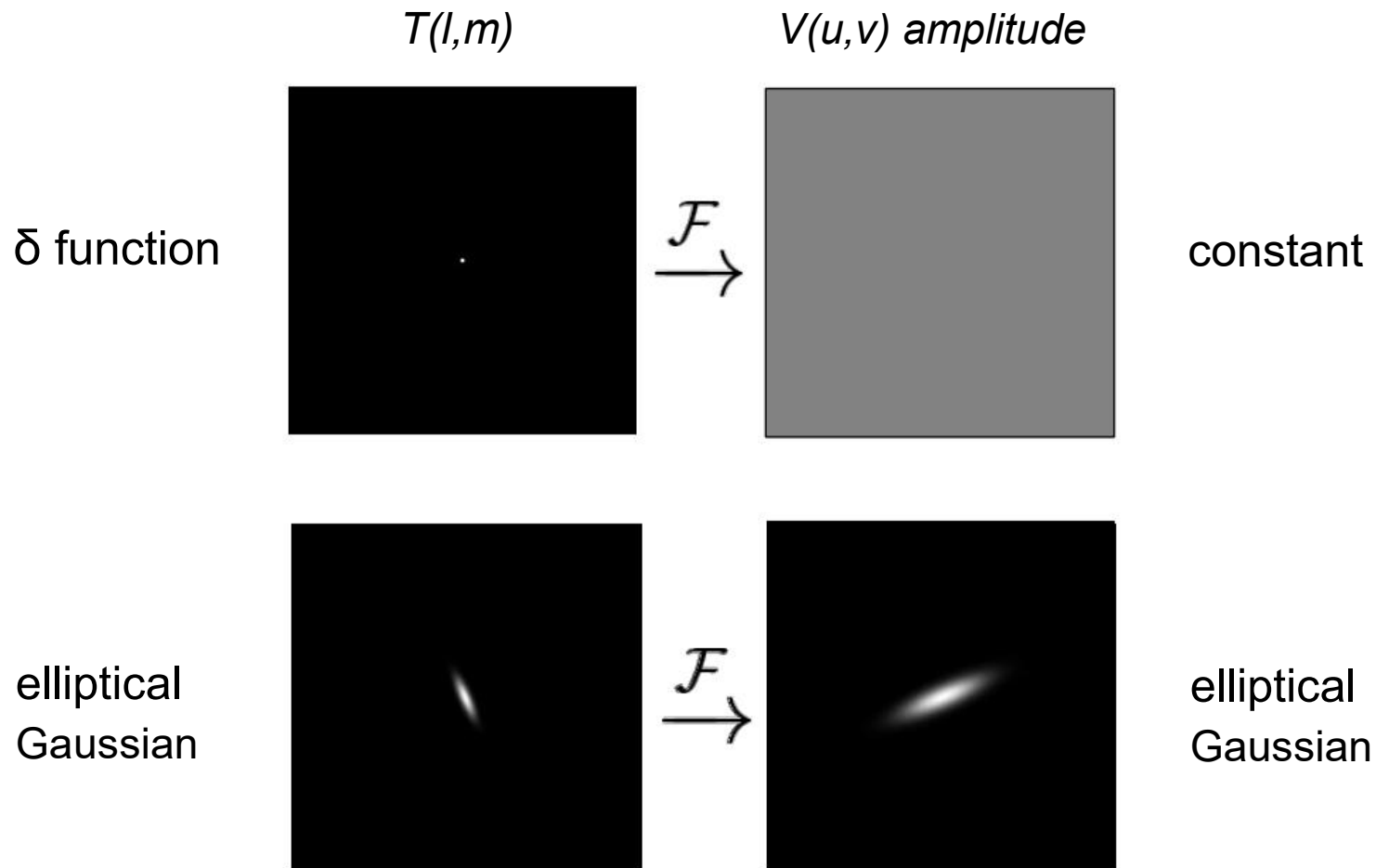


Visibilities

- each $V(u,v)$ contains information on $T(l,m)$ *everywhere*, not just at a given (l,m) coordinate or within a particular subregion
- each $V(u,v)$ is a complex quantity
 - expressed as (*real, imaginary*) or (*amplitude, phase*)



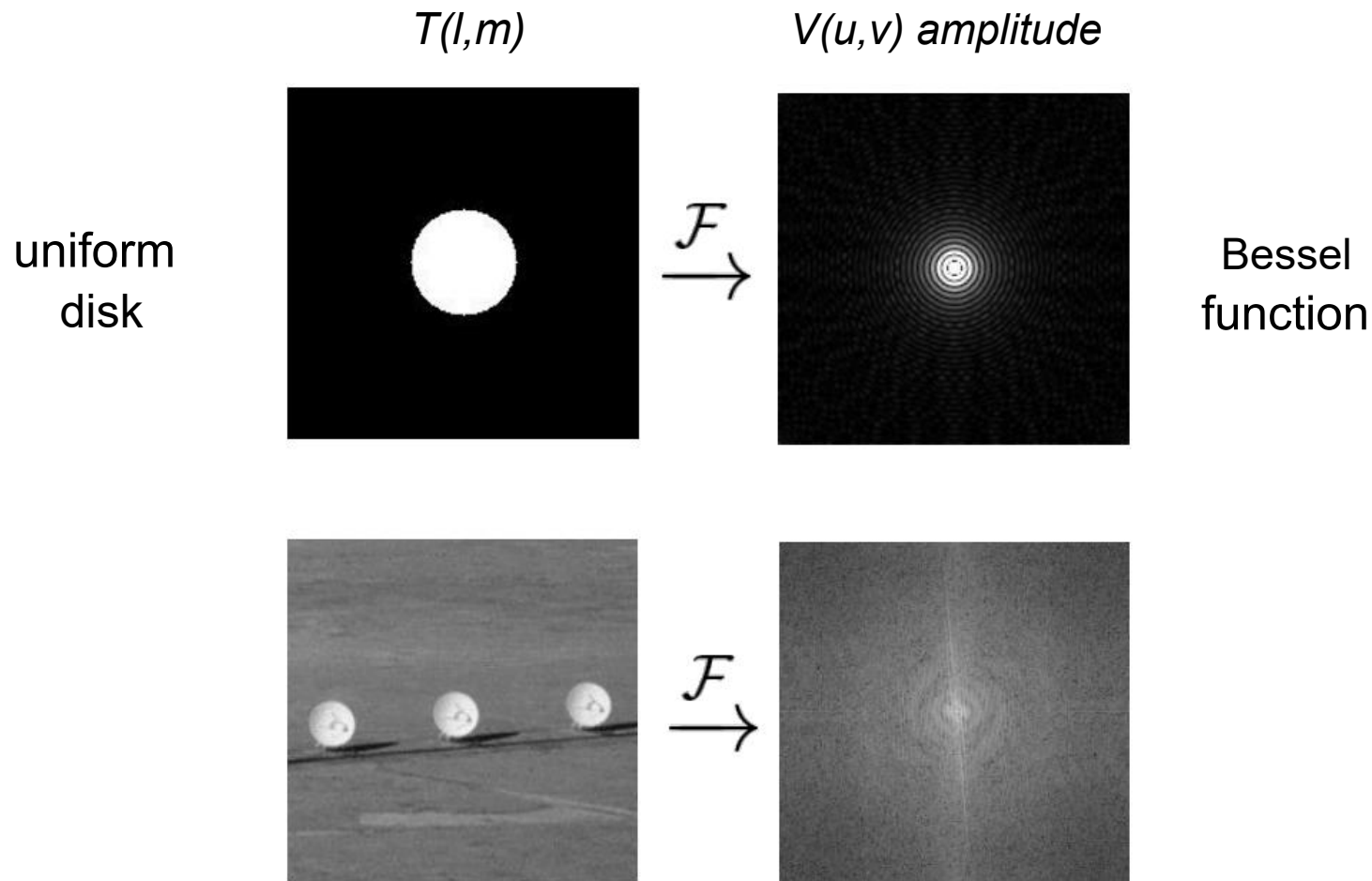
Example 2D Fourier Transforms



narrow features transform into wide features (and vice-versa)



Example 2D Fourier Transforms

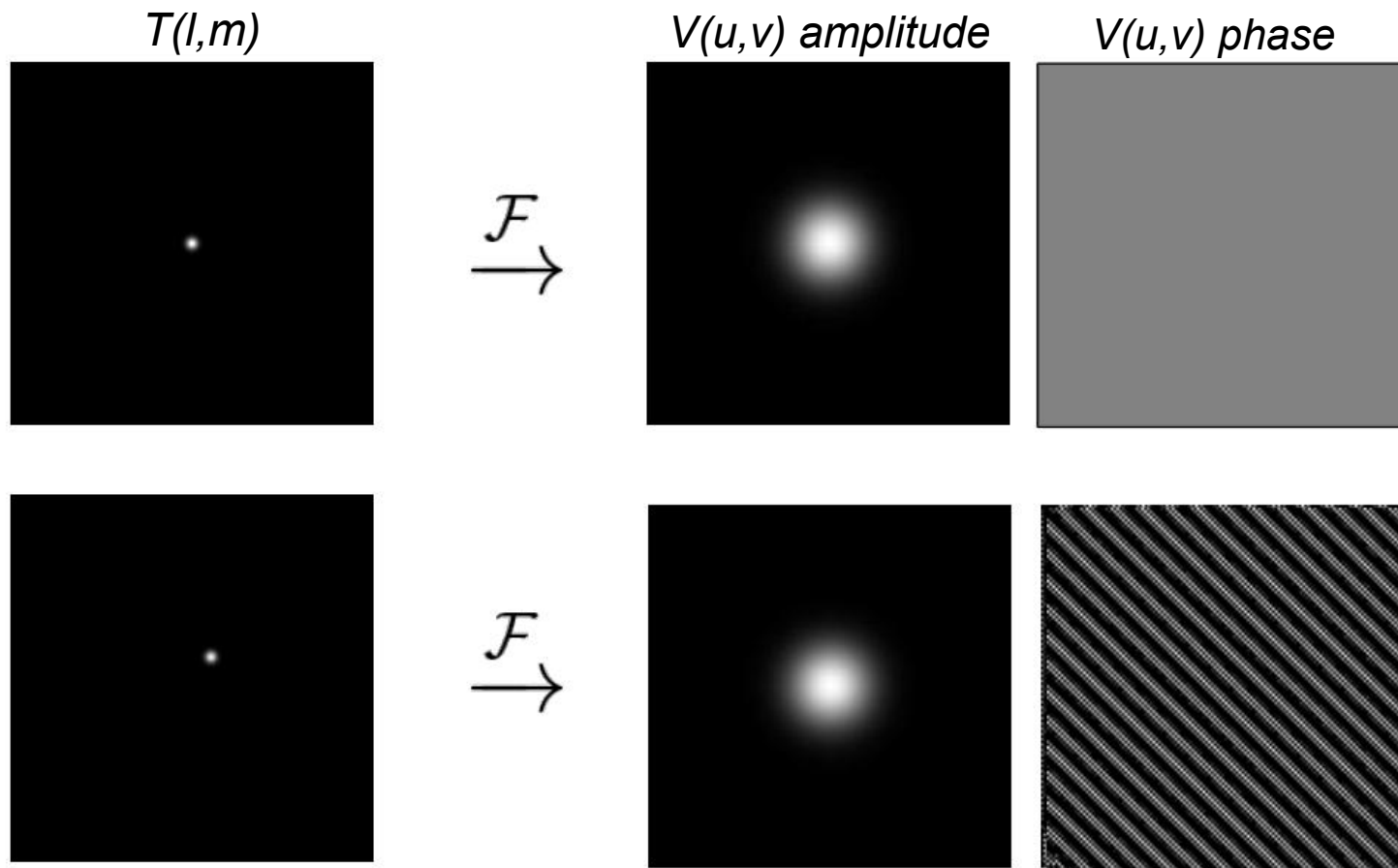


sharp edges result in many high spatial frequencies



Amplitude and Phase

- amplitude tells “how much” of a certain spatial frequency
- phase tells “where” this spatial frequency component is located



The Visibility Concept

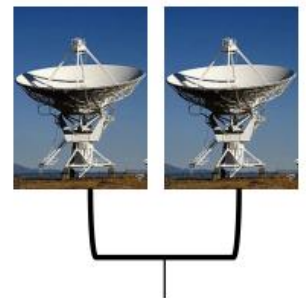
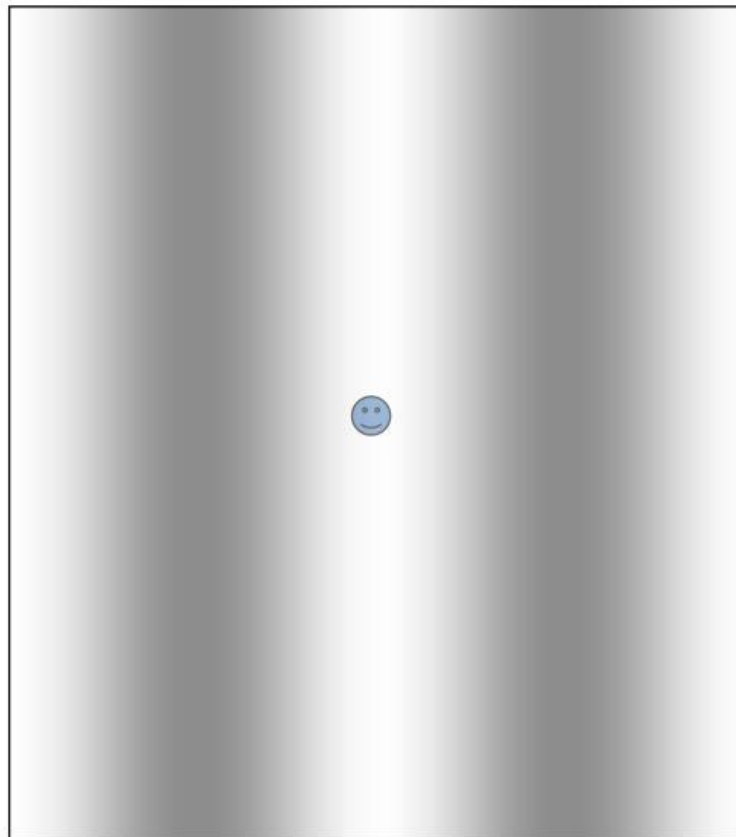
$$V(u, v) = \int \int T(l, m) e^{-i2\pi(ul+vm)} dl dm$$

- visibility as a function of baseline coordinates (u, v) is the Fourier transform of the sky brightness distribution as a function of the sky coordinates (l, m)
- $V(u=0, v=0)$ is the integral of $T(l, m) dl dm =$ total flux density
- since $T(l, m)$ is real, $V(-u, -v) = V^*(u, v)$
 - $V(u, v)$ is Hermitian
 - get two visibilities for one measurement



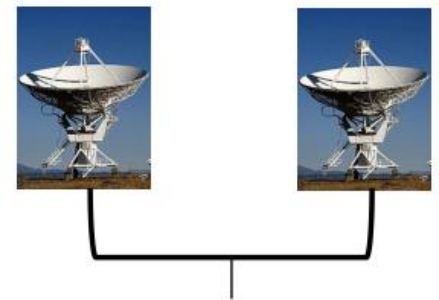
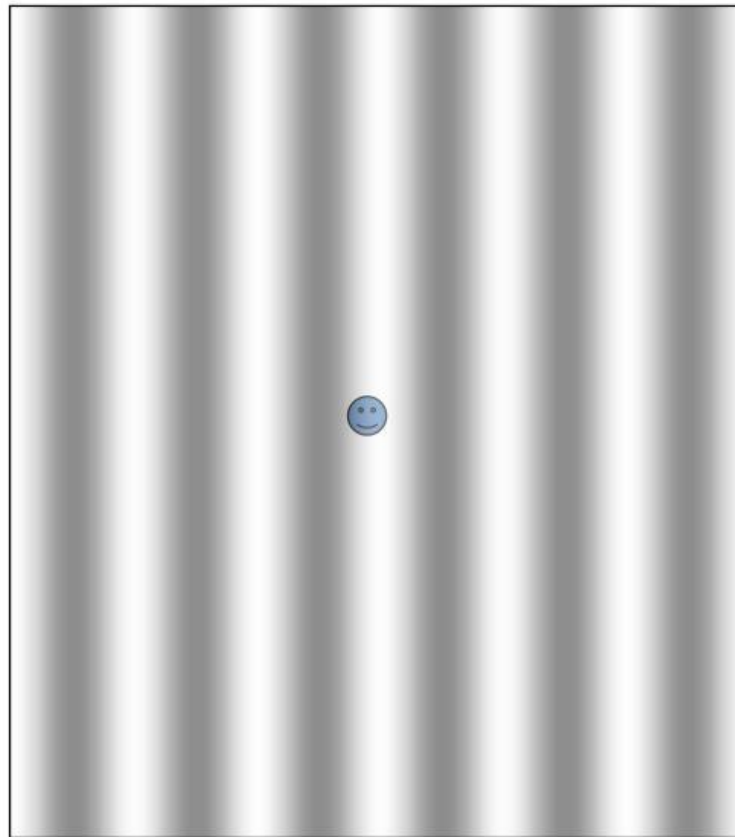
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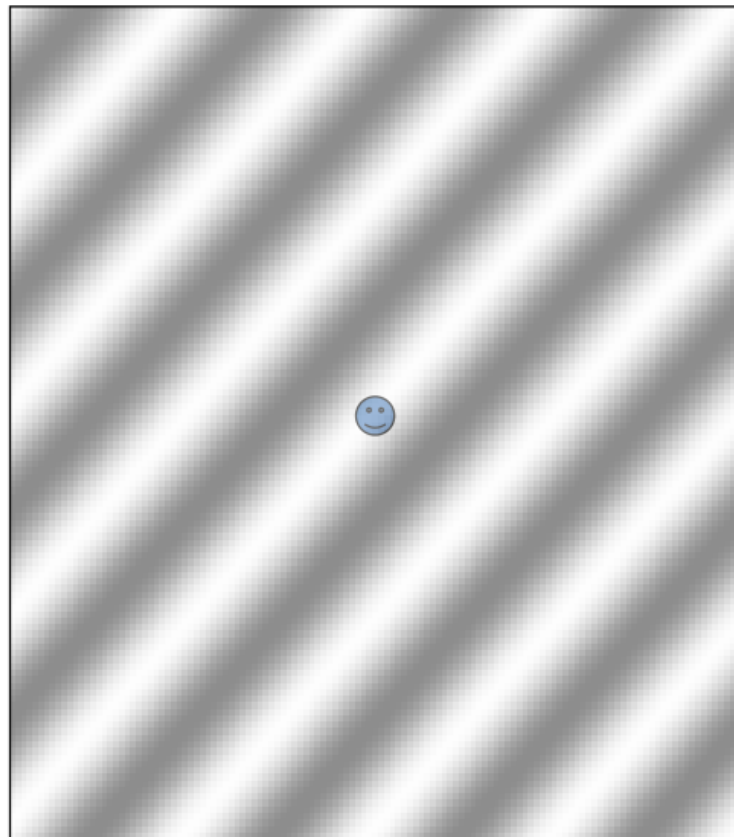
The Visibility Concept

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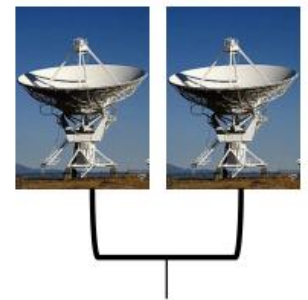
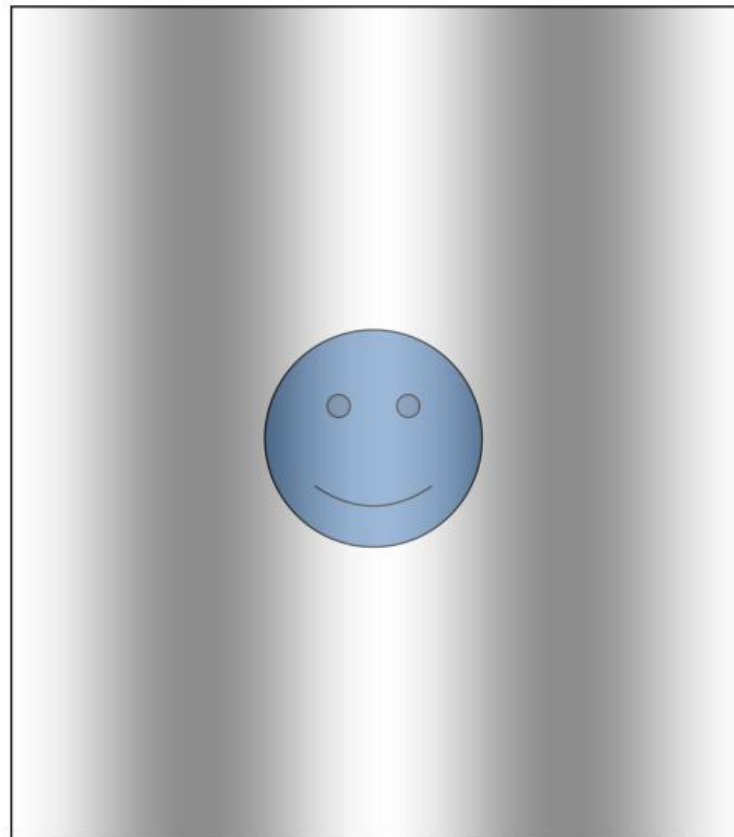
The Visibility Concept

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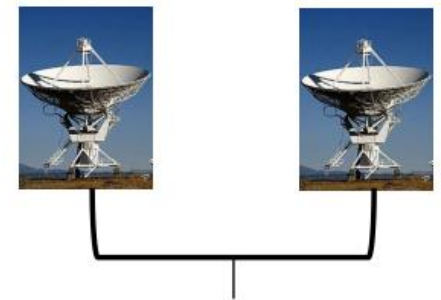
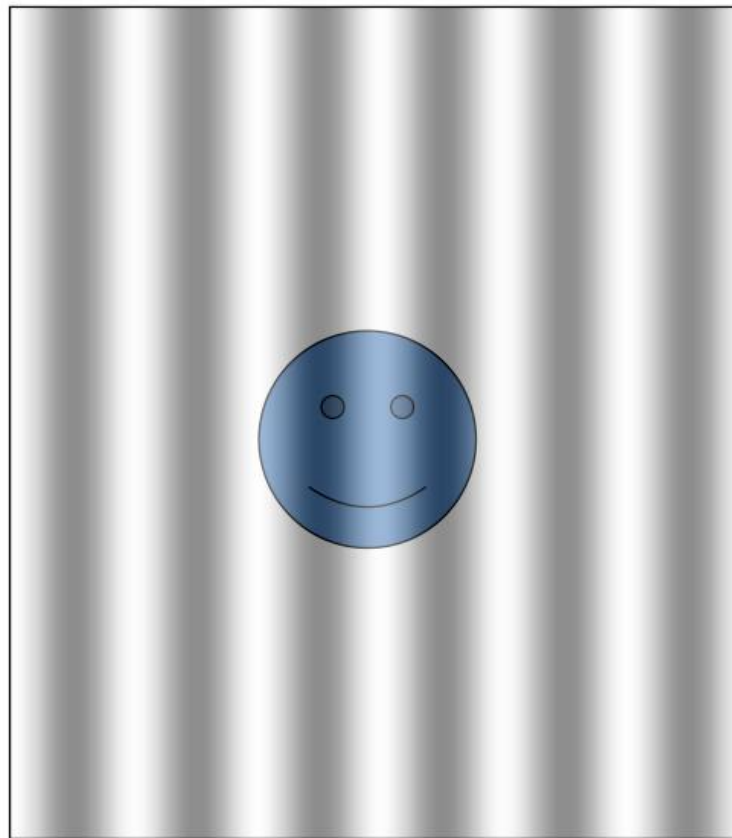
The Visibility Concept

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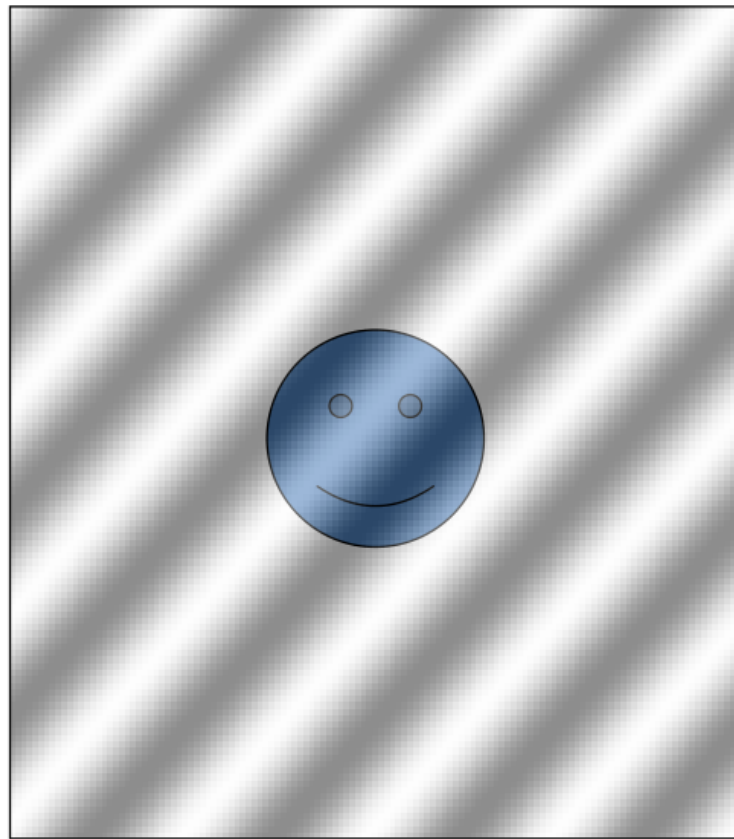
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The Visibility Concept

$$V(u, v) = \iint T(l, m) e^{-i2\pi(ul+vm)} dl dm$$



Aperture Synthesis Basics

- idea: sample $V(u,v)$ at enough (u,v) points using distributed small aperture antennas to synthesize a large aperture antenna of size (u_{max}, v_{max})
- one pair of antennas = one baseline
= two (u,v) samples at a time
- N antennas = $N(N-1)$ samples at a time
- use Earth rotation to fill in (u,v) plane over time
(Sir Martin Ryle, 1974 Nobel Prize in Physics)
- reconfigure physical layout of N antennas for more samples
- observe at multiple wavelengths for (u,v) plane coverage, for source spectra amenable to simple characterization (“multi-frequency synthesis”)
- if source is variable, then be careful



Sir Martin Ryle
1918-1984

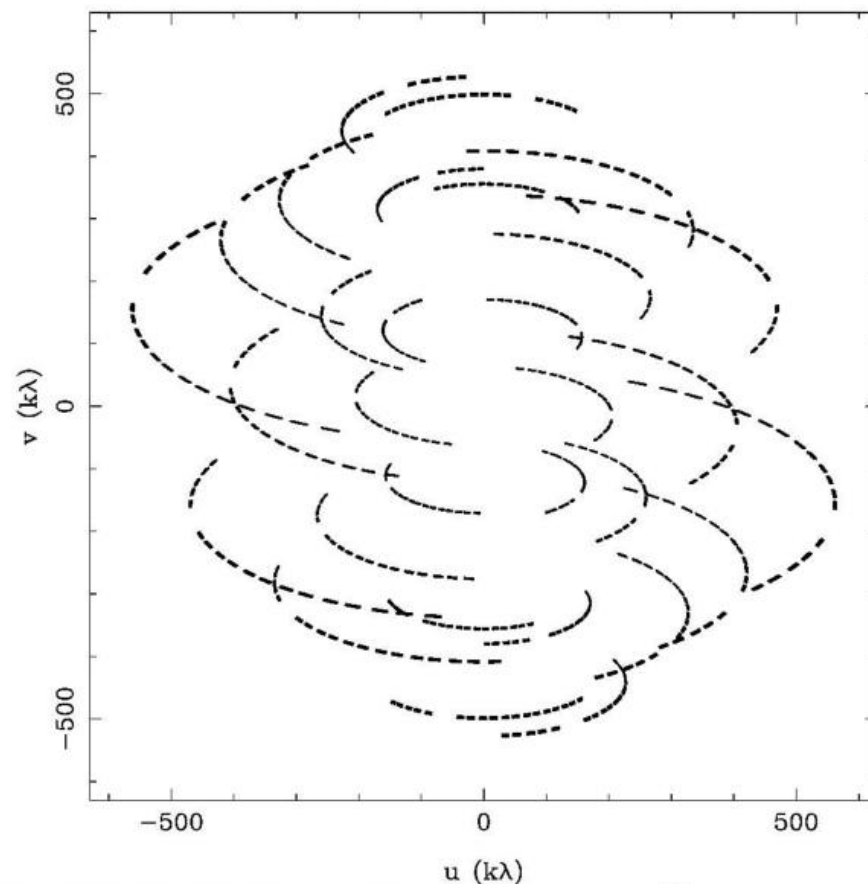
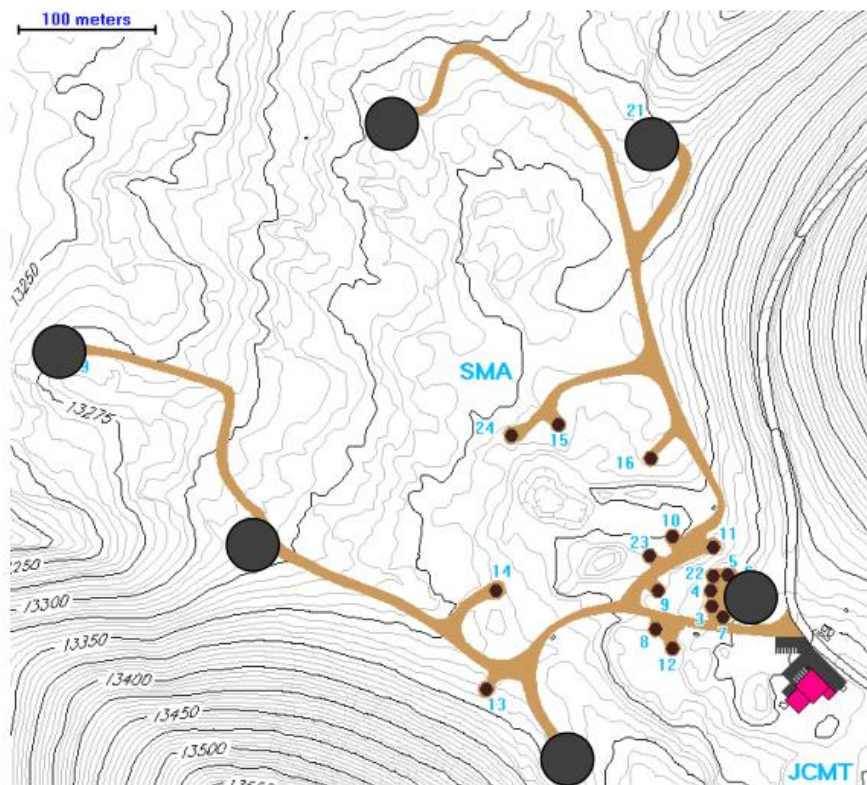


Examples of Aperture Synthesis Telescopes (for Millimeter Wavelengths)



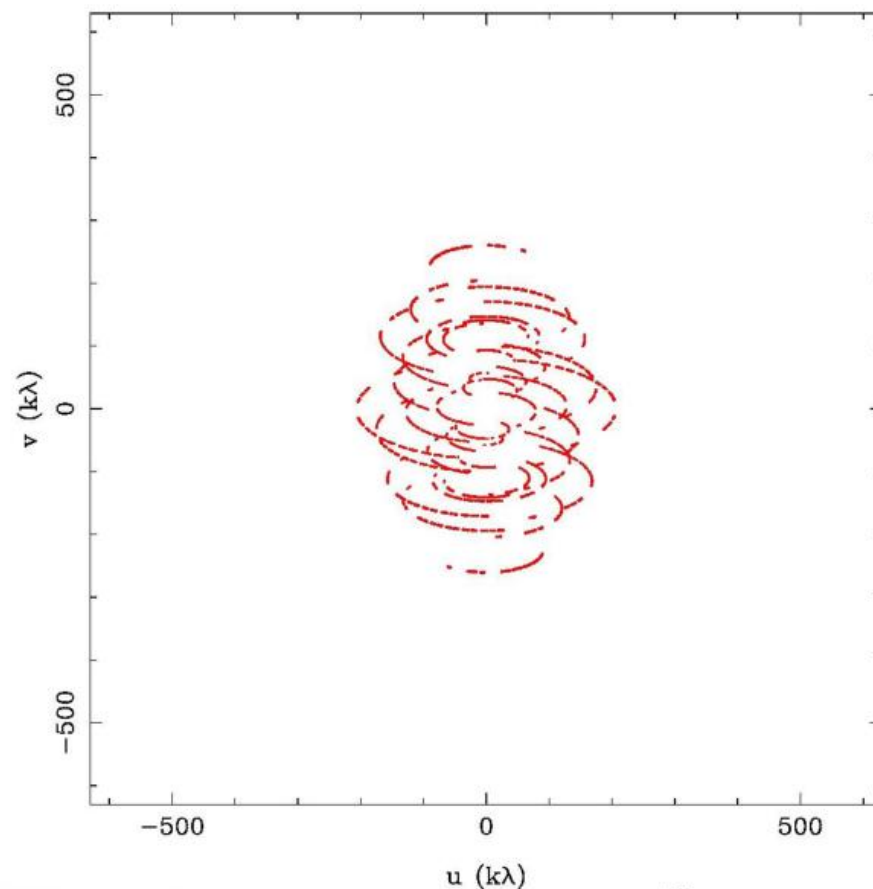
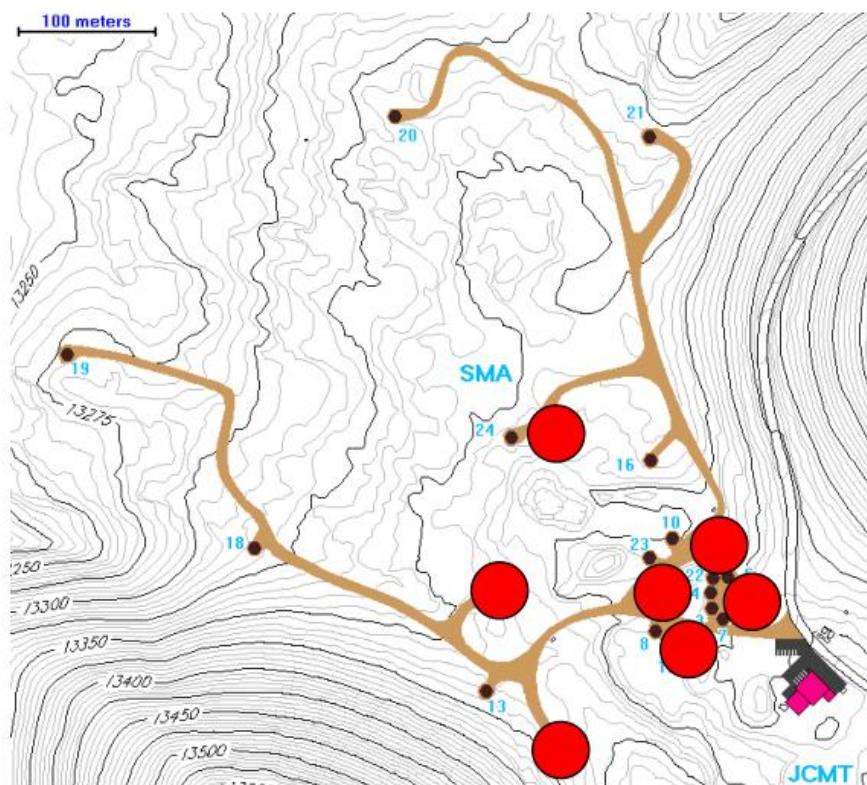
An Example of (u,v) plane Sampling

VEX configuration of 6 SMA antennas, $\nu = 345$ GHz, $\text{dec} = +22^\circ$



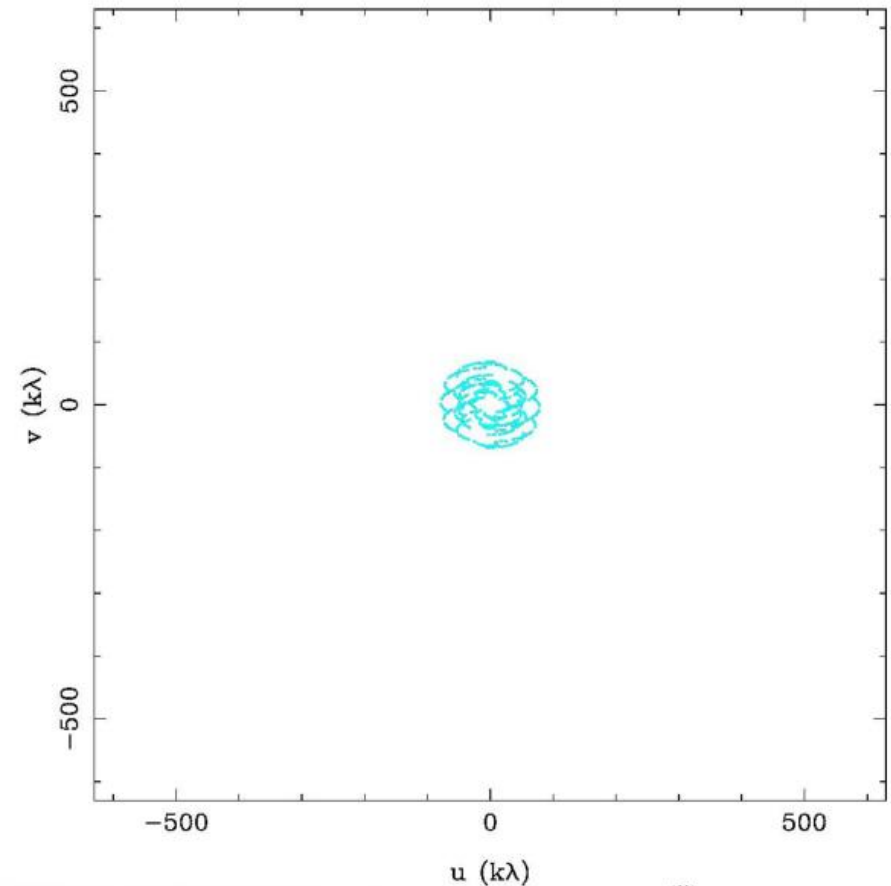
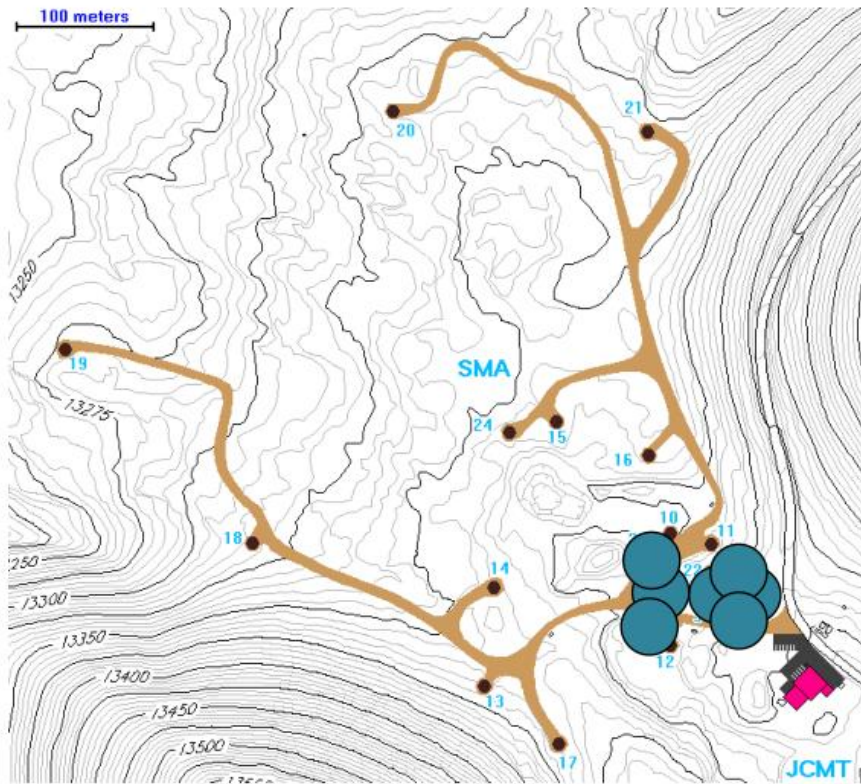
An Example of (u,v) plane Sampling

EXT configurations of 7 SMA antennas, $\nu = 345$ GHz, $\text{dec} = +22^\circ$



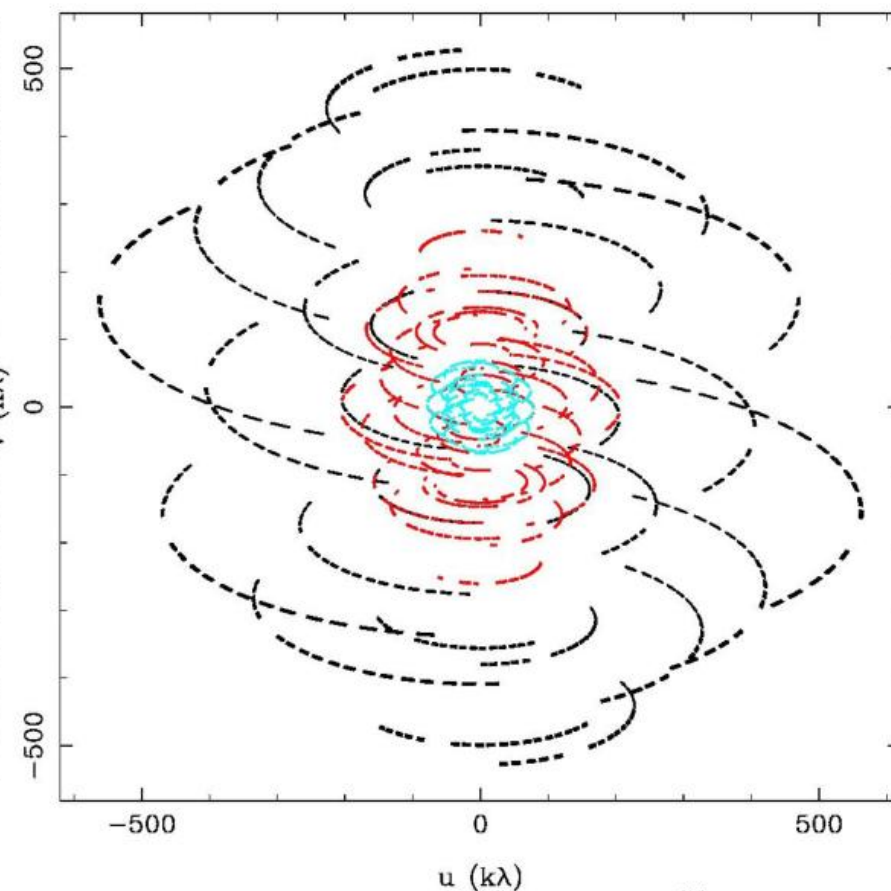
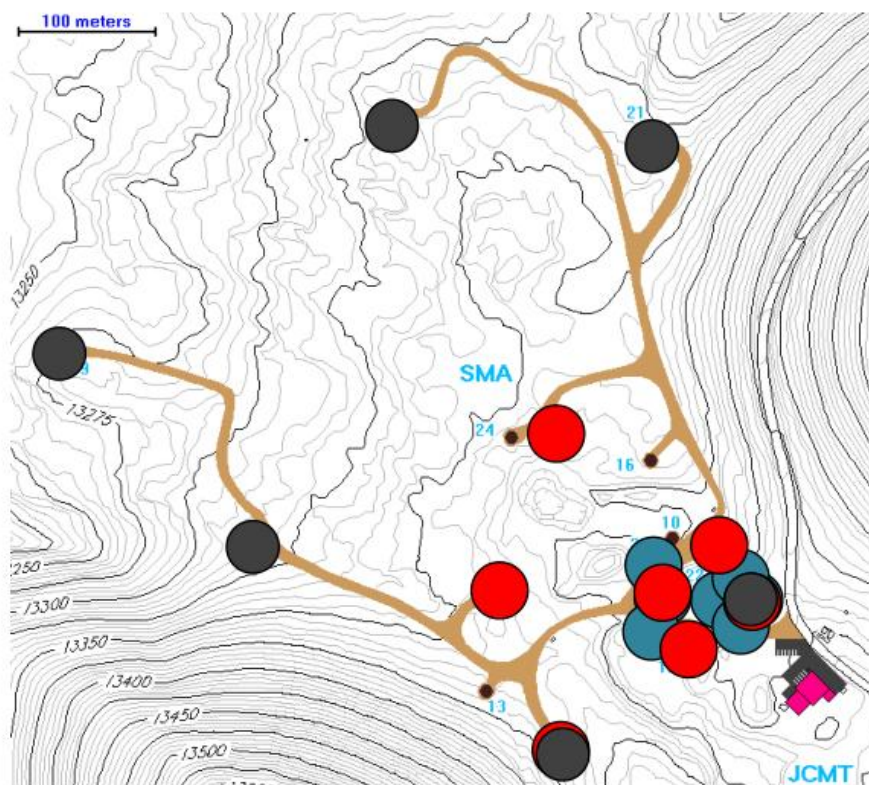
An Example of (u,v) plane Sampling

COM configurations of 7 SMA antennas, $\nu = 345$ GHz, $\text{dec} = +22^\circ$



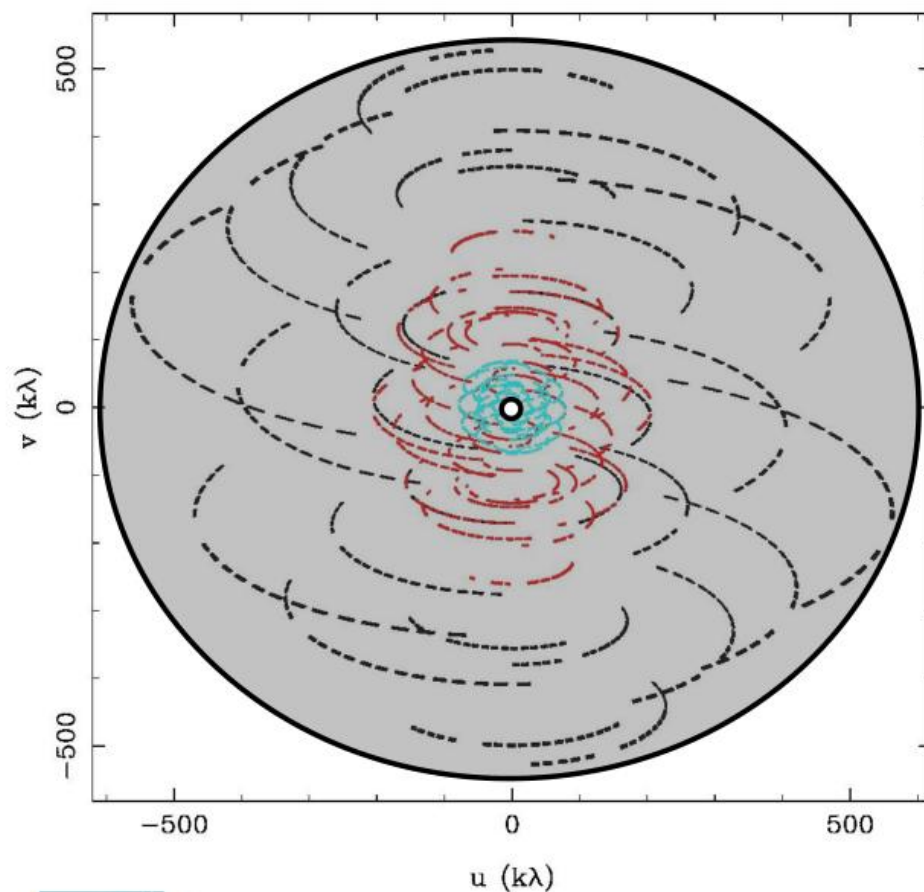
An Example of (u,v) plane Sampling

3 configurations of SMA antennas, $\nu = 345$ GHz, dec = +22 deg



Implications of (u,v) plane Sampling

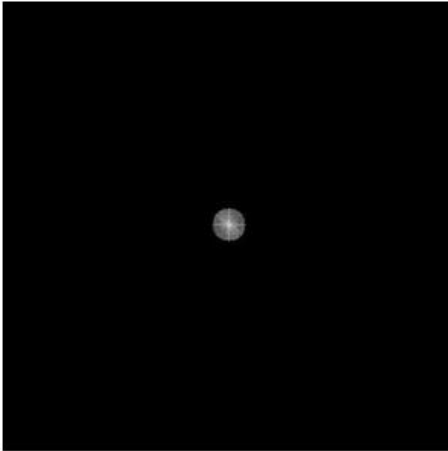
samples of $V(u,v)$ are limited by number of antennas and by Earth-sky geometry



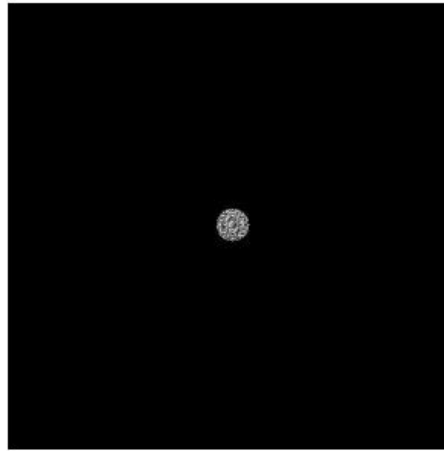
- *outer boundary*
 - no information on smaller scales
 - resolution limit
- *inner hole*
 - no information on larger scales
 - extended structures invisible
- *irregular coverage between boundaries*
 - sampling theorem violated
 - information missing

Inner and Outer (u,v) Boundaries

$V(u,v)$ amplitude



$V(u,v)$ phase

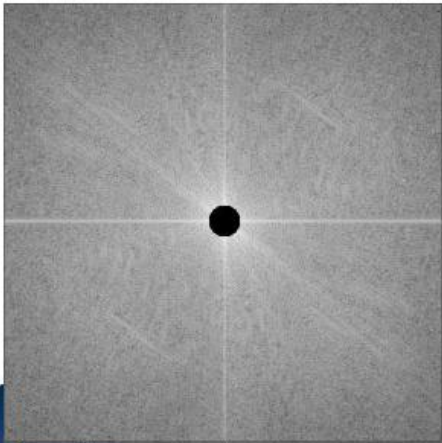


$\mathcal{F}$
 $\rightarrow$

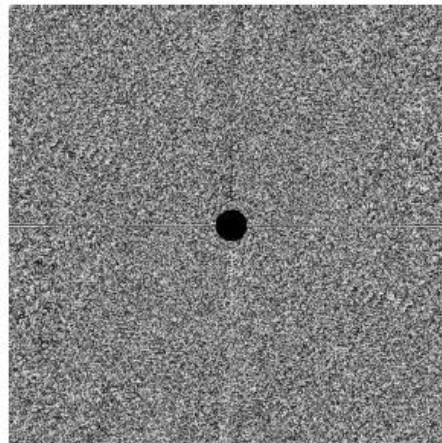
$T(l,m)$



$V(u,v)$ amplitude



$V(u,v)$ phase

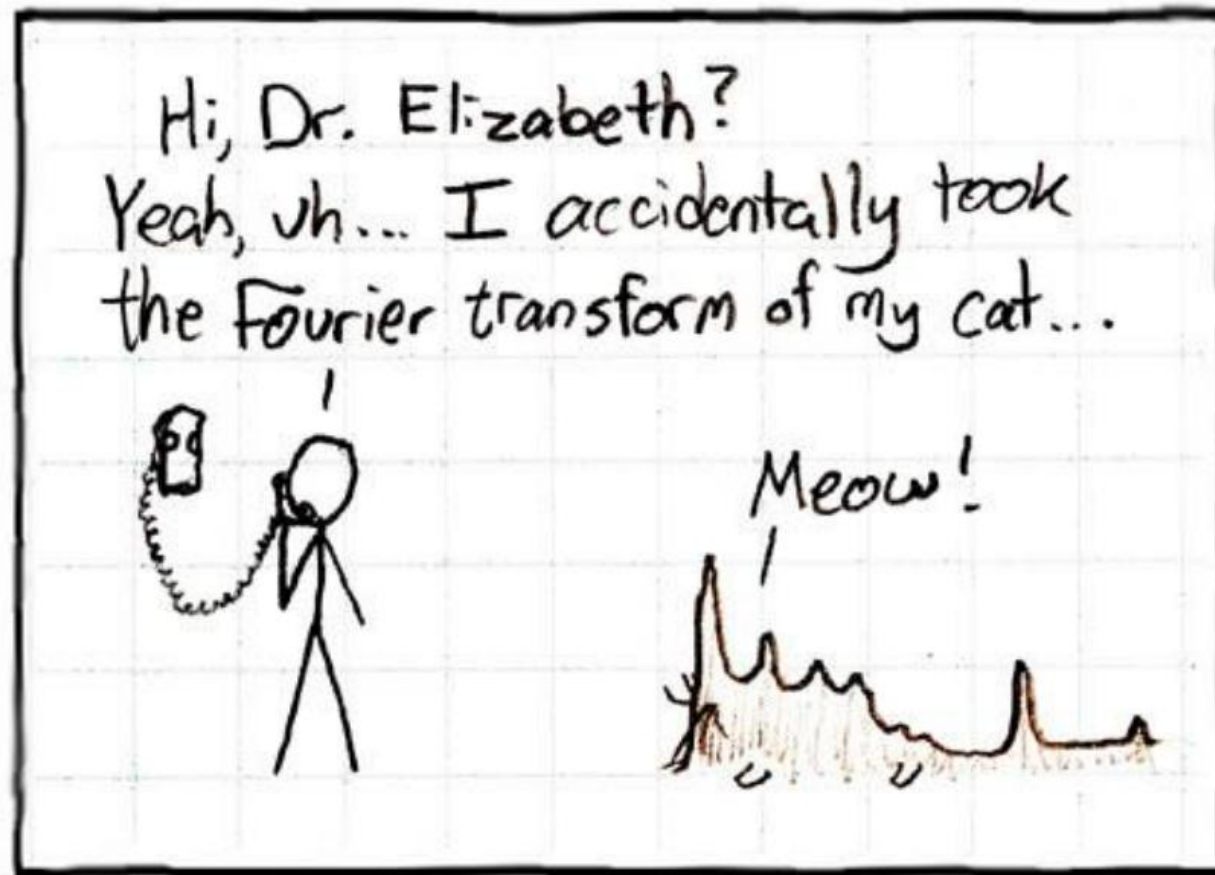


$\mathcal{F}$
 $\rightarrow$

$T(l,m)$



xkcd.com/26/



Formal Description of Imaging

$$V(u, v) \xrightarrow{\mathcal{F}} T(l, m)$$

- sample Fourier domain at discrete points $S(u, v) = \sum_{k=1}^M \delta(u - u_k, v - v_k)$
- Fourier transform sampled visibility function $V(u, v)S(u, v) \xrightarrow{\mathcal{F}} T^D(l, m)$
- apply the convolution theorem $T(l, m) * s(l, m) = T^D(l, m)$

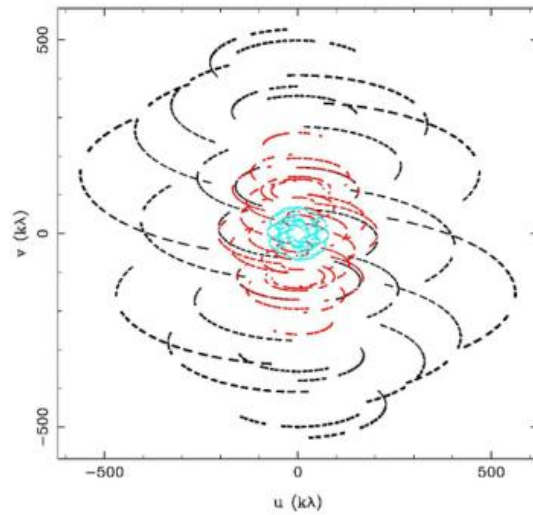
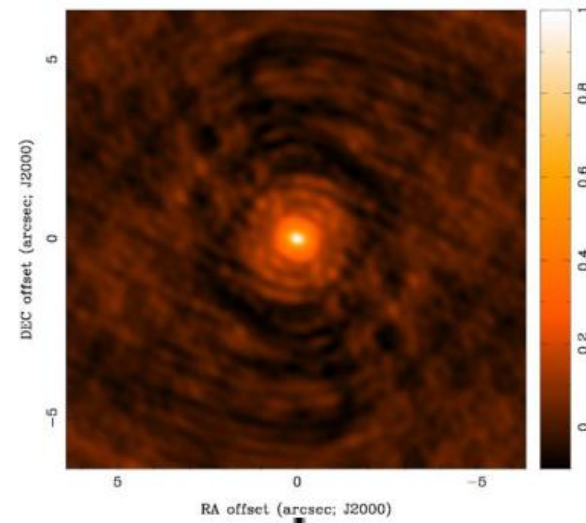
where the Fourier transform of the sampling pattern $s(l, m) \xrightarrow{\mathcal{F}} S(u, v)$ is the “point spread function”

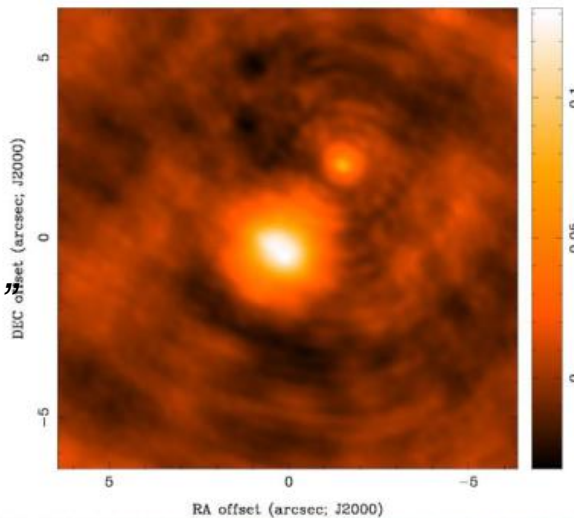
the Fourier transform of the sampled visibilities yields the true sky brightness convolved with the point spread function

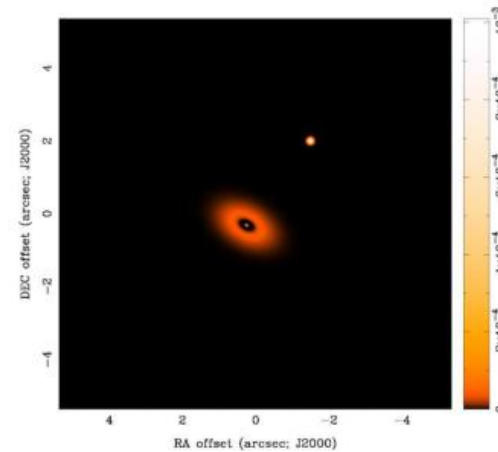
radio jargon: the “dirty image” is the true image convolved with the “dirty beam”



Dirty Beam and Dirty Image

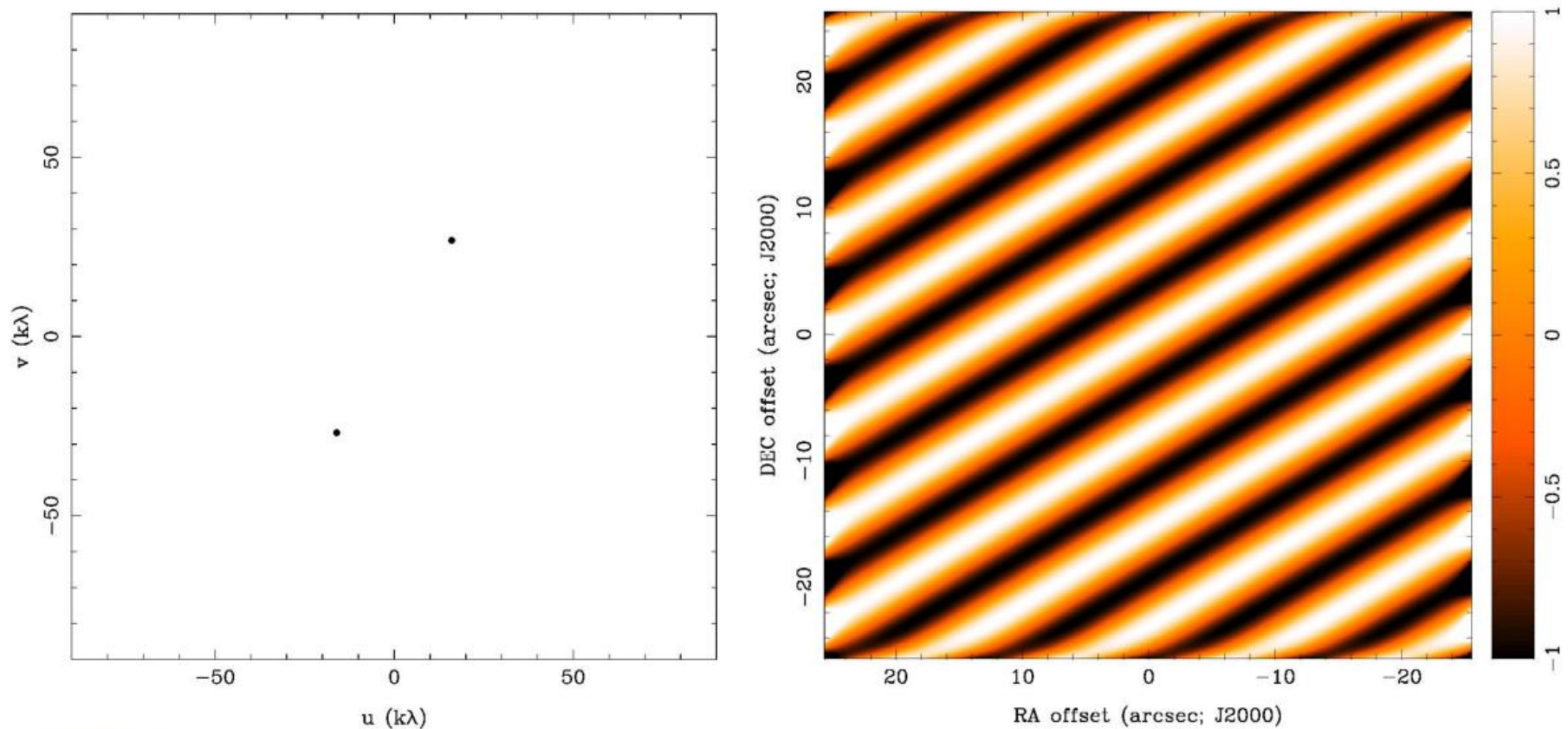
 $S(u, v)$

 $\mathcal{F}$

 $s(l, m)$
 “dirty beam”

 $*$

 $T^D(l, m)$
 “dirty image”

 $\leftarrow$

 $T(l, m)$

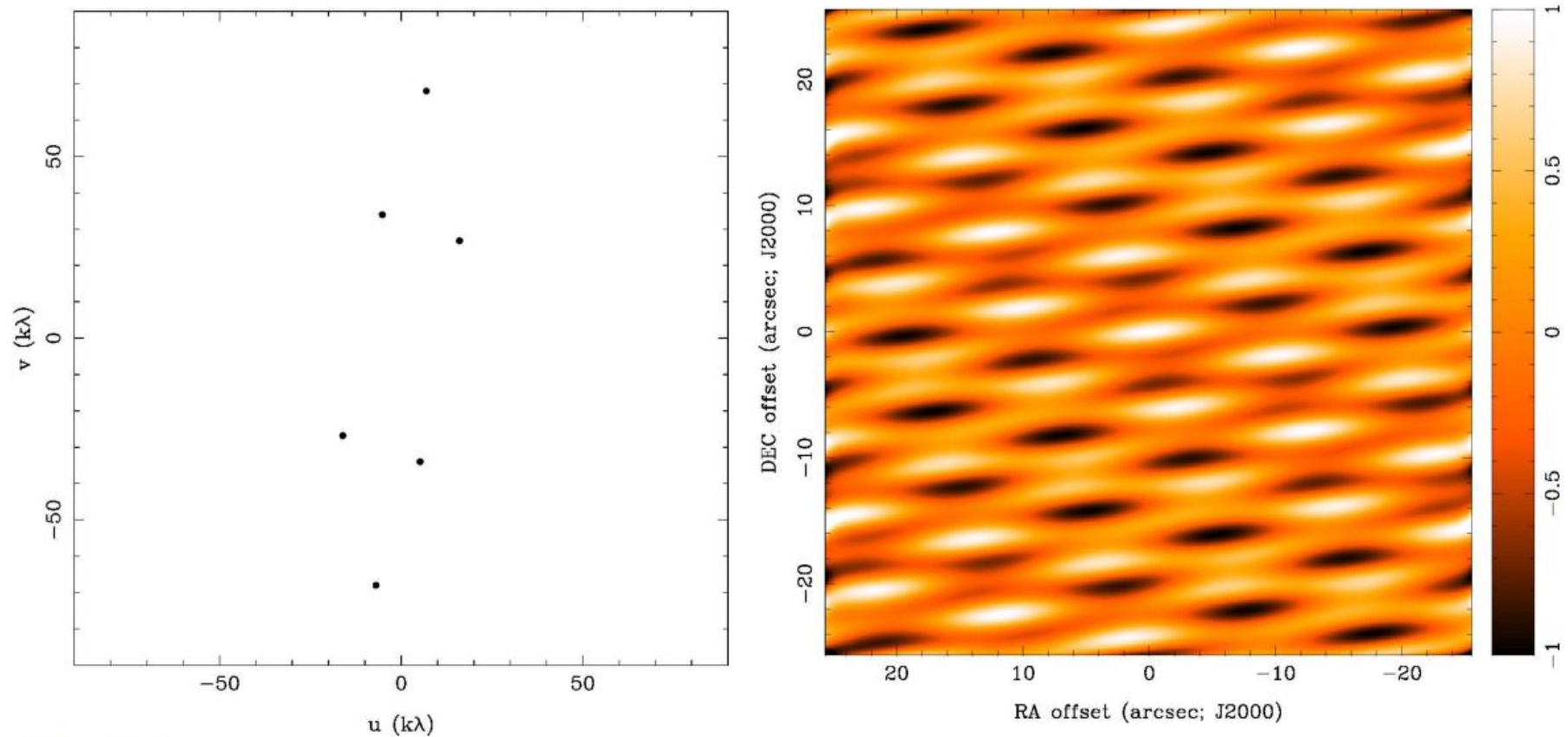

Dirty Beam Shape and N Antennas

2 Antennas, 1 Sample



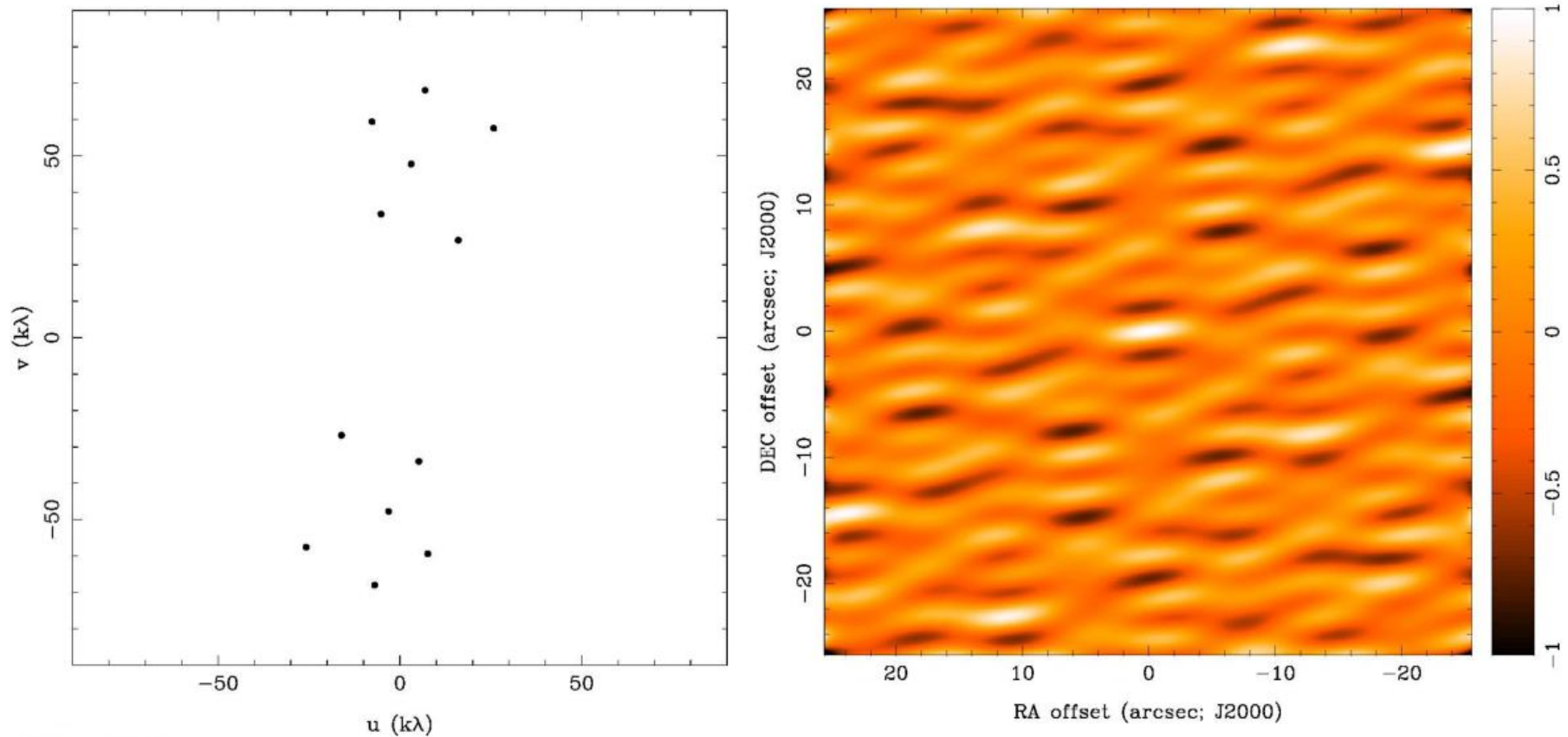
Dirty Beam Shape and N Antennas

3 Antennas, 1 Sample



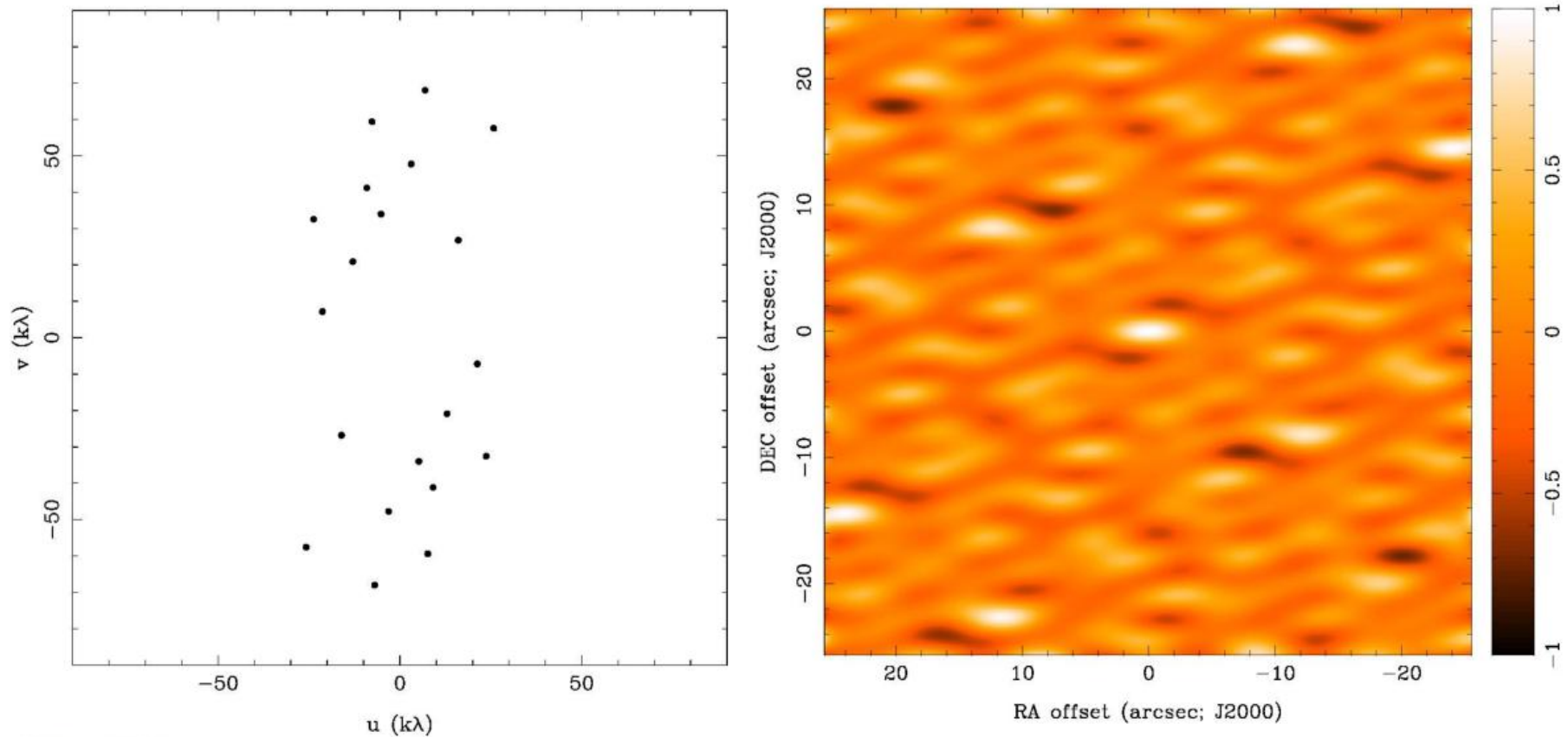
Dirty Beam Shape and N Antennas

4 Antennas, 1 Sample



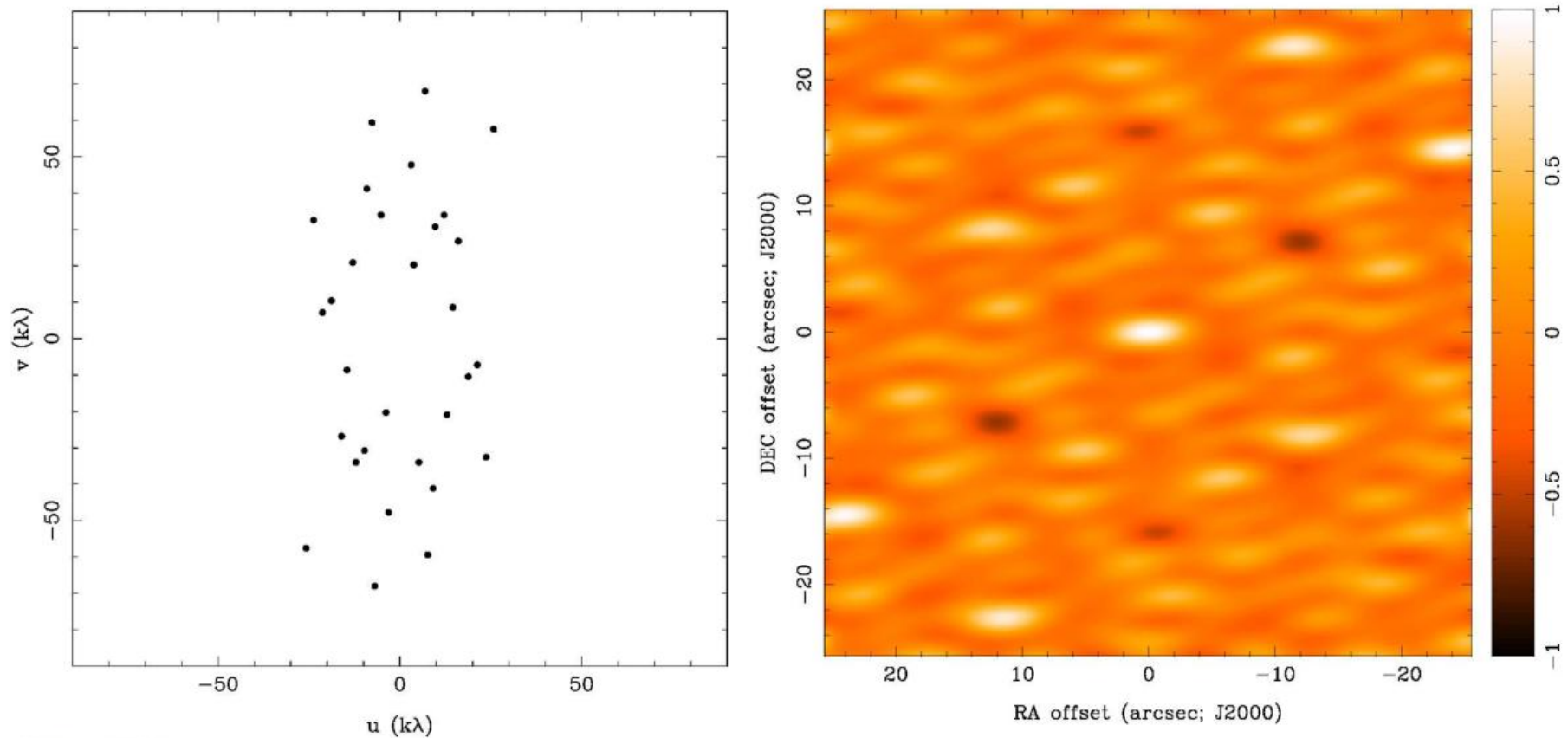
Dirty Beam Shape and N Antennas

5 Antennas, 1 Sample



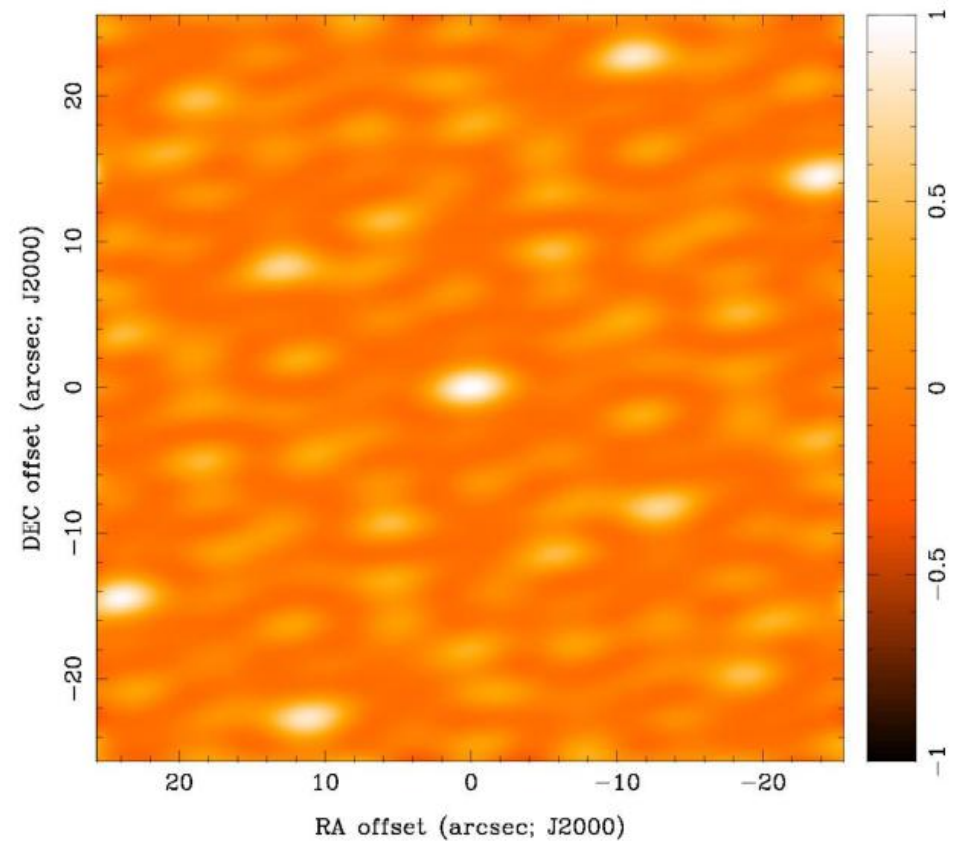
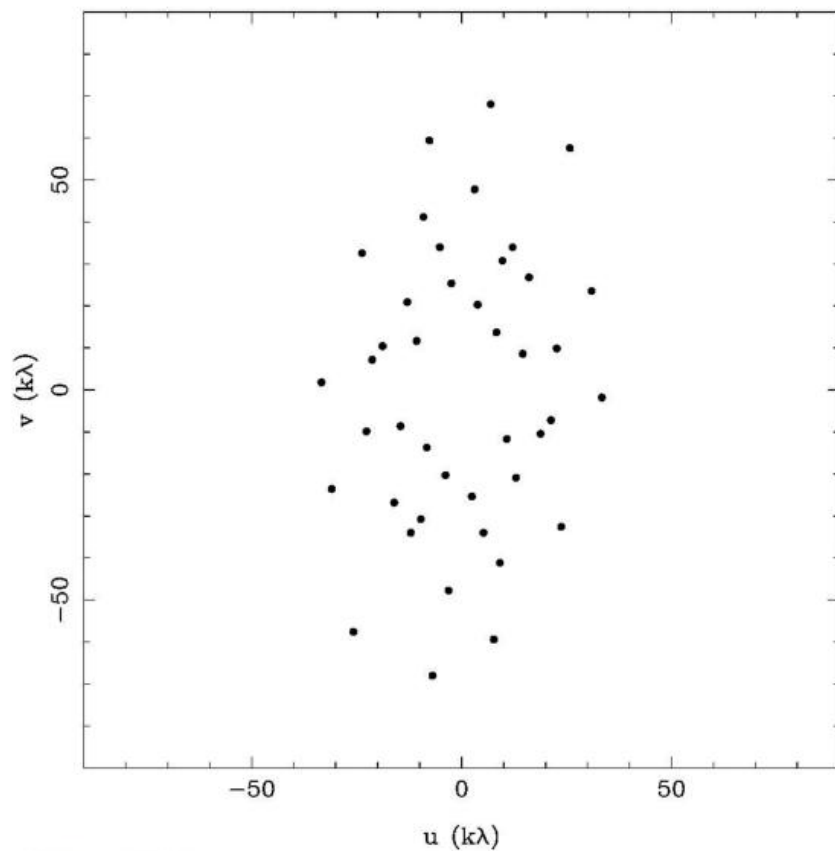
Dirty Beam Shape and N Antennas

6 Antennas, 1 Sample



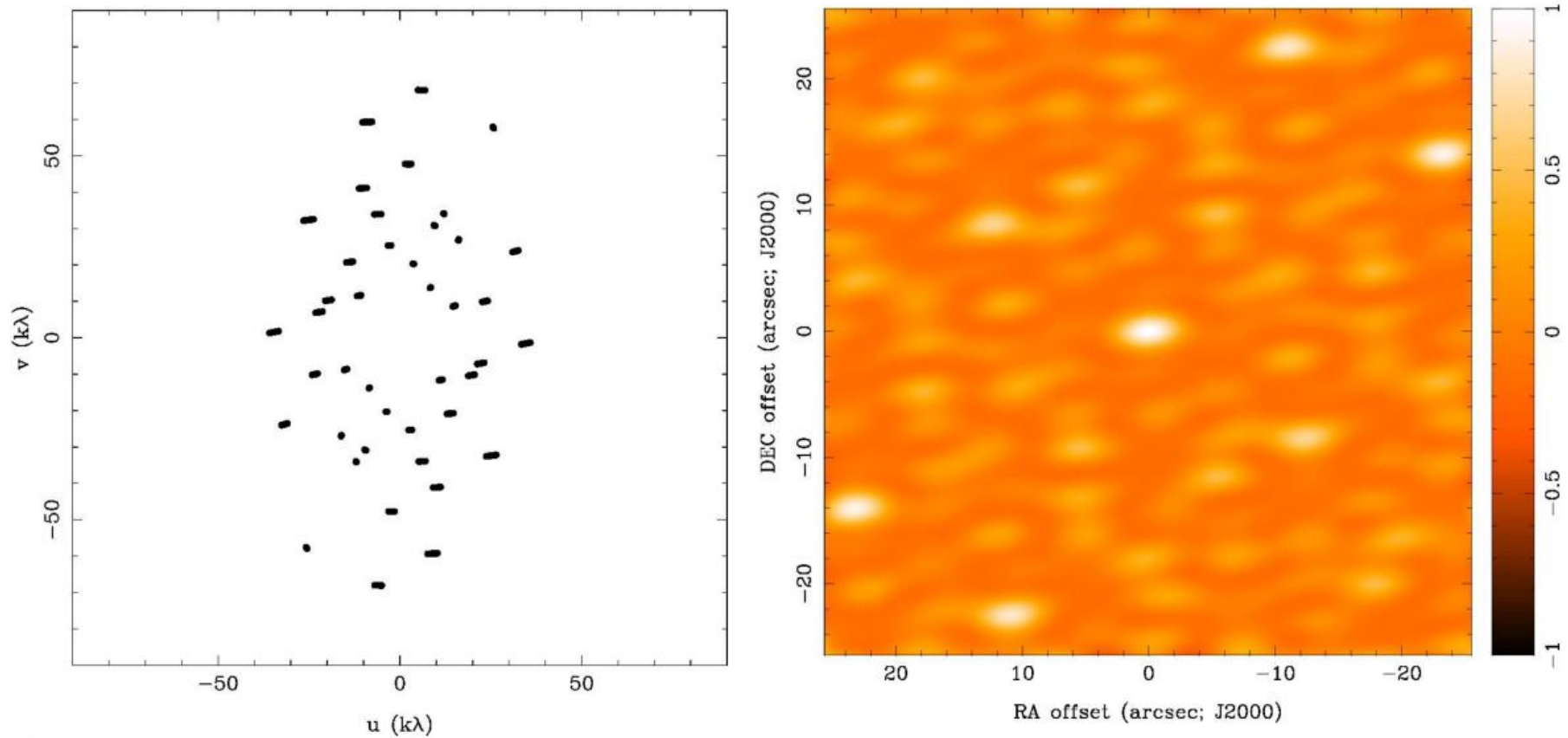
Dirty Beam Shape and N Antennas

7 Antennas, 1 Sample



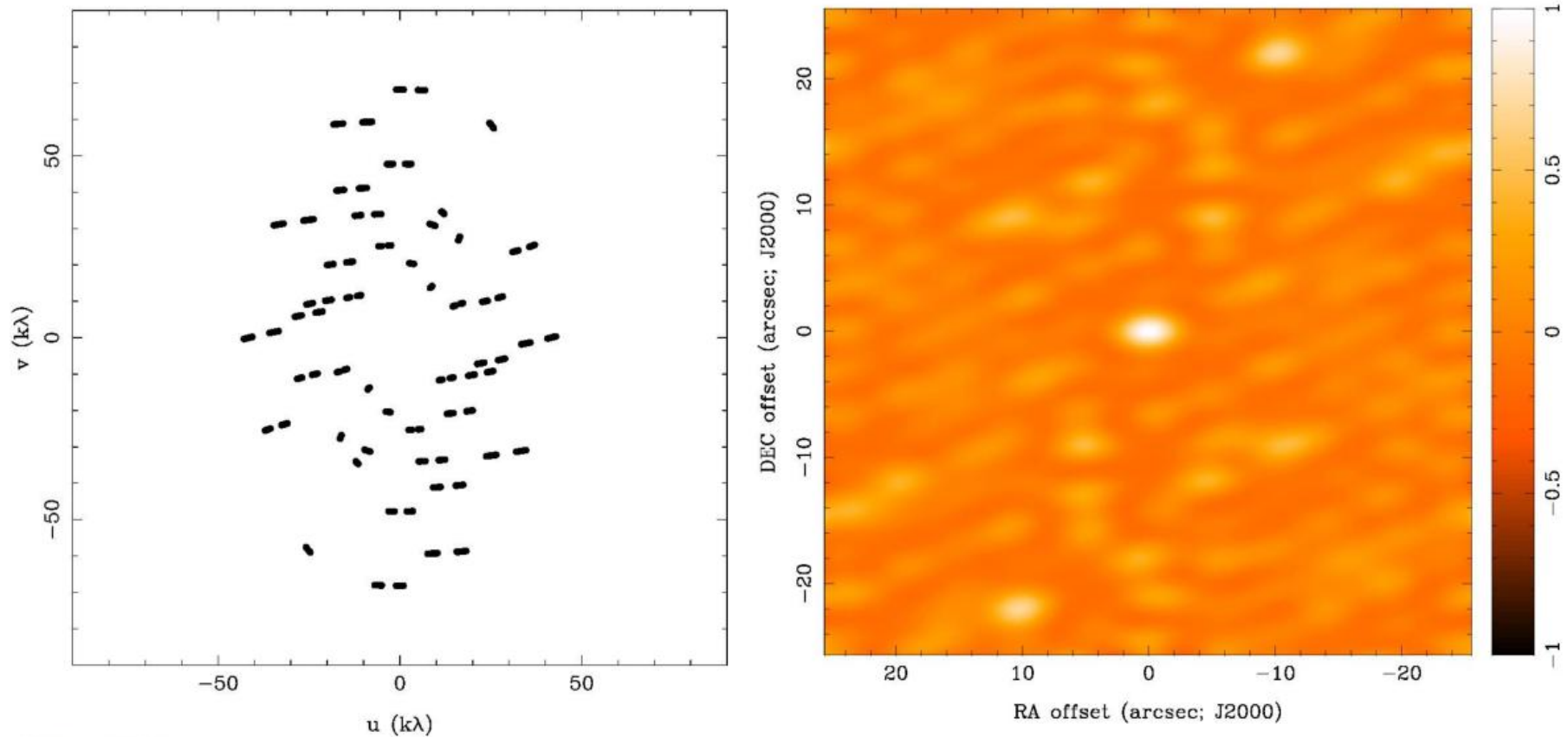
Dirty Beam Shape and N Antennas

7 Antennas, 10 min



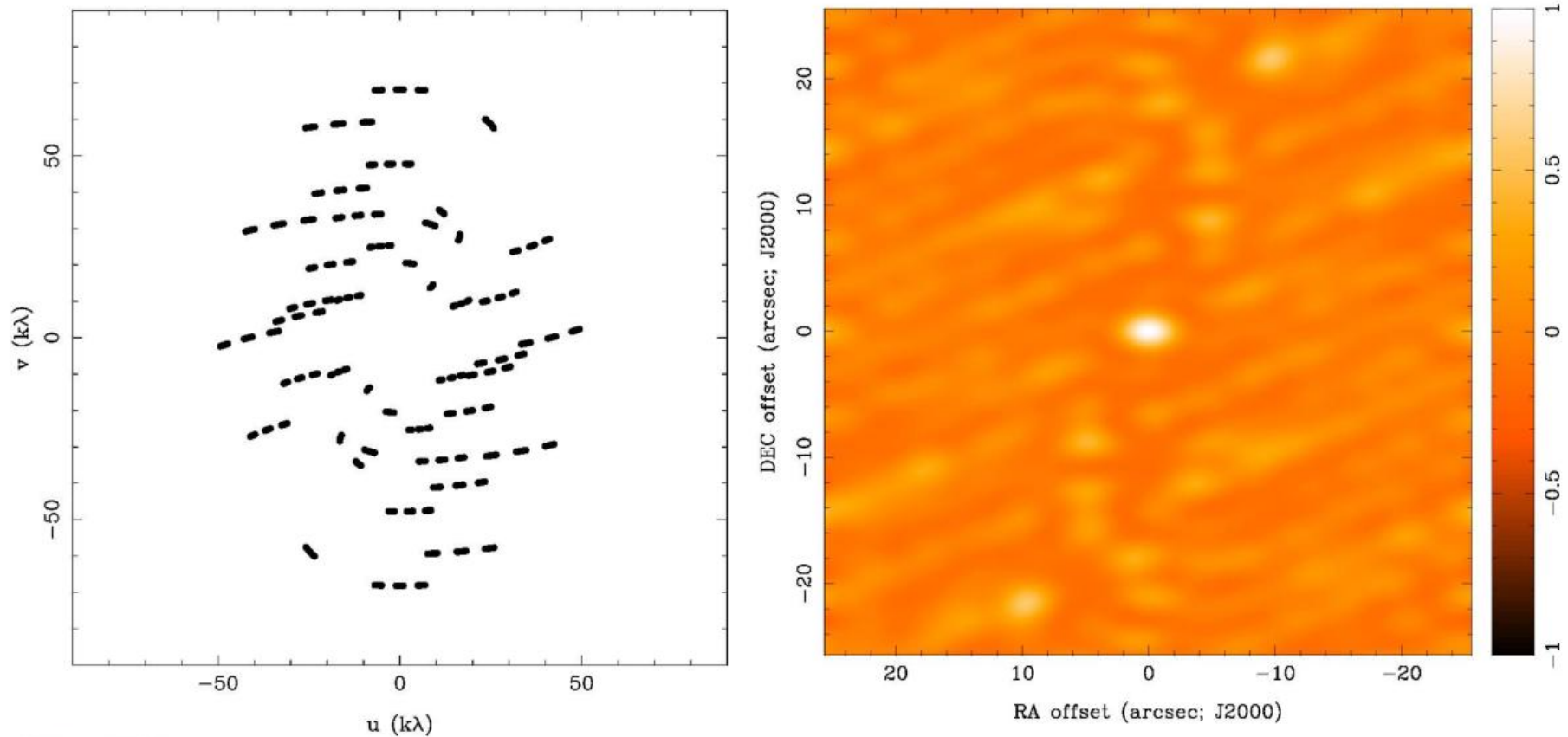
Dirty Beam Shape and N Antennas

7 Antennas, 2 x 10 min



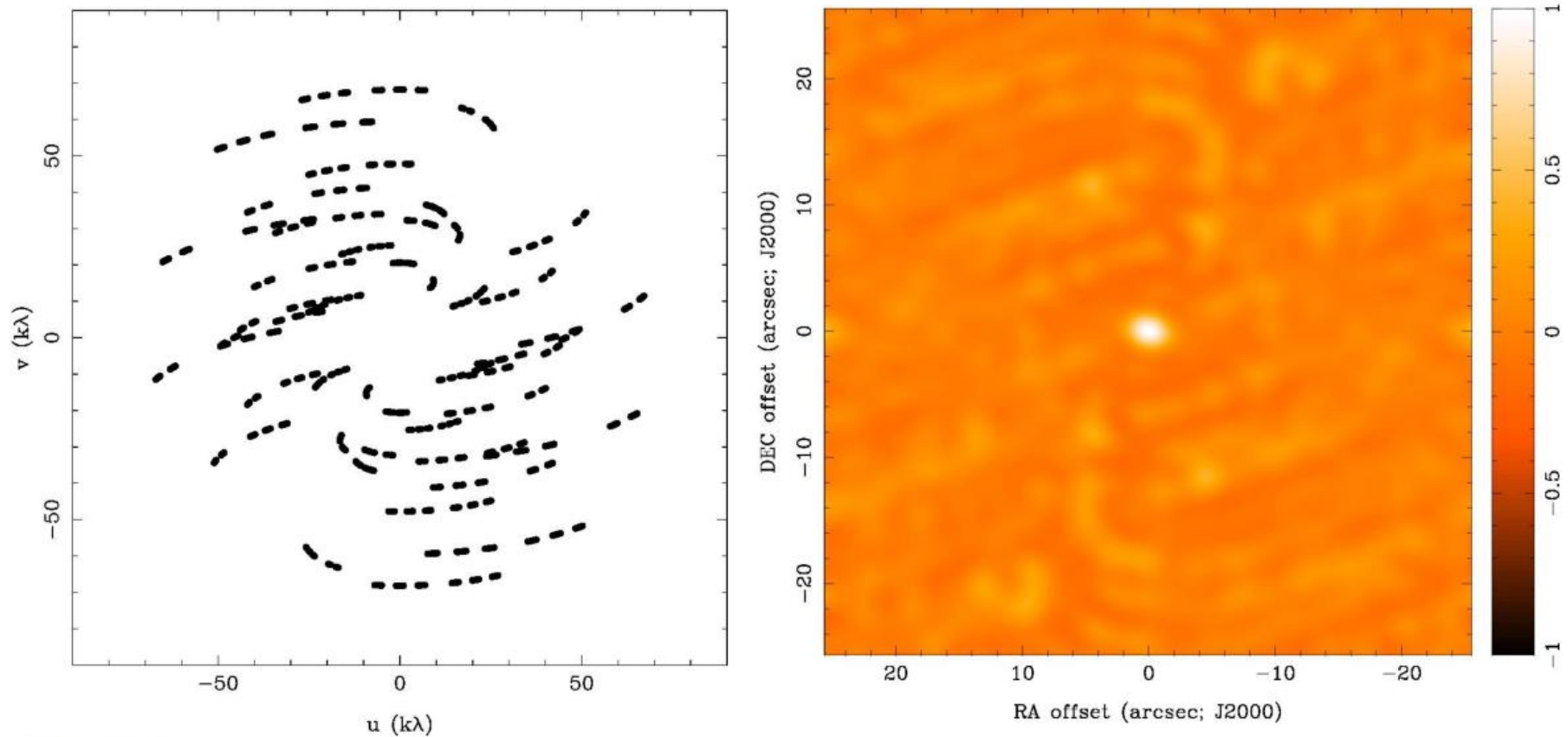
Dirty Beam Shape and N Antennas

7 Antennas, 1 hour



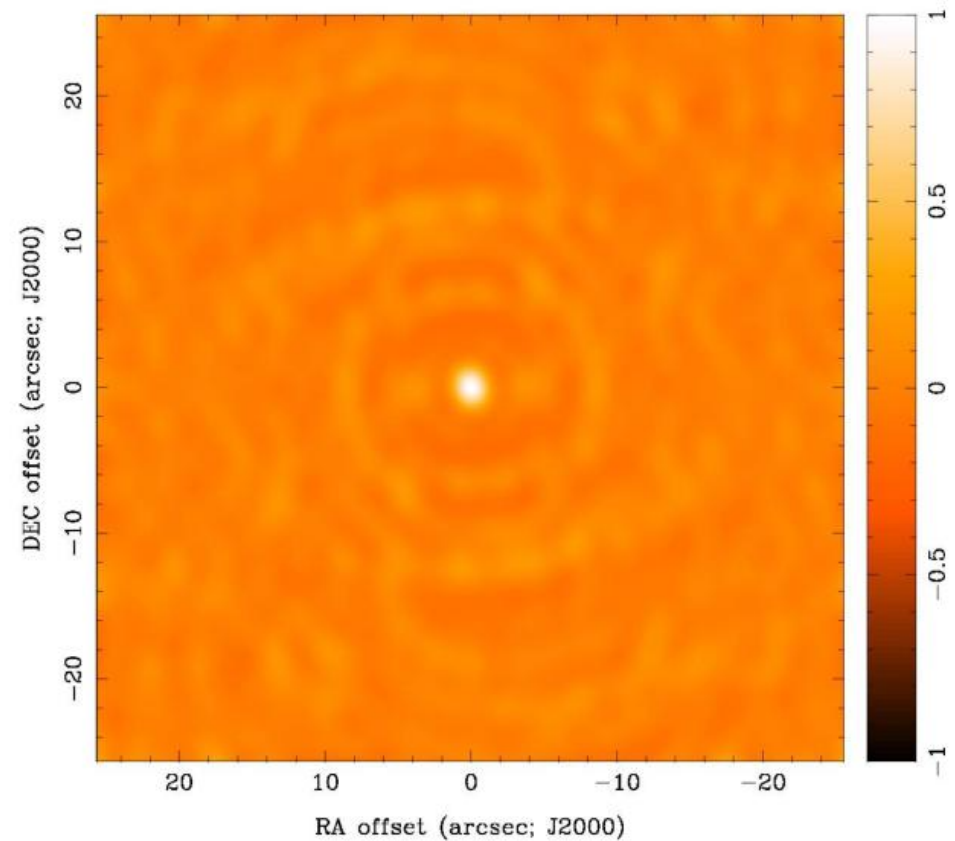
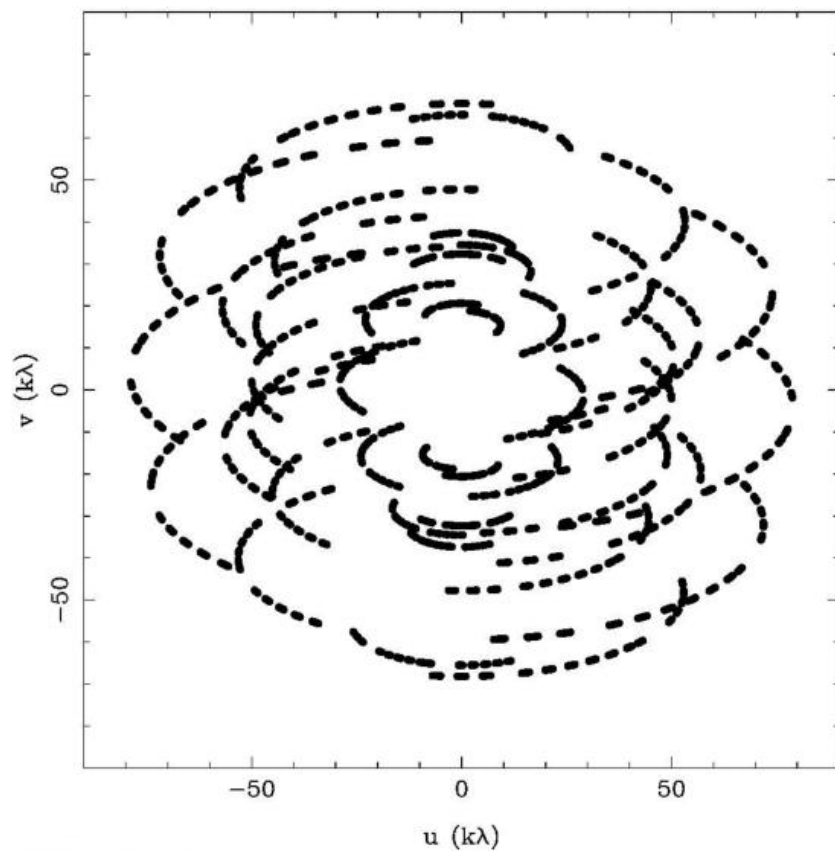
Dirty Beam Shape and N Antennas

7 Antennas, 3 hours



Dirty Beam Shape and N Antennas

7 Antennas, 8 hours



Calibrated Visibilities: What's Next?

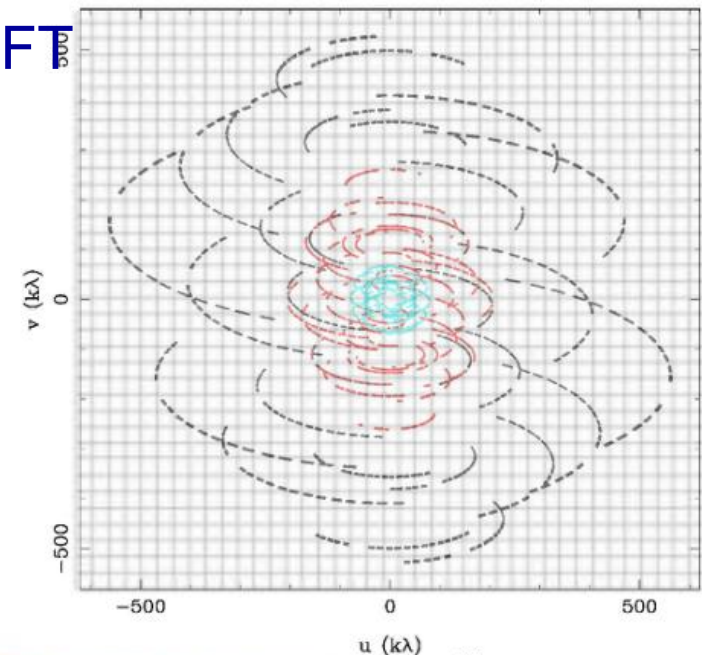
- analyze directly $V(u,v)$ samples by model fitting
 - good for simple structures, e.g. point sources, symmetric disks
 - sometimes for statistical descriptions of sky brightness
 - visibilities have very well defined noise properties
- recover an image from the observed incomplete and noisy samples of its Fourier transform for analysis
 - Fourier transform $V(u,v)$ to get $T^D(l,m)$
 - difficult to do science with the dirty image $T^D(l,m)$
 - deconvolve $s(l,m)$ from $T^D(l,m)$ to determine a model of $T(l,m)$
 - work with the model of $T(l,m)$



Some Details of the Dirty Image

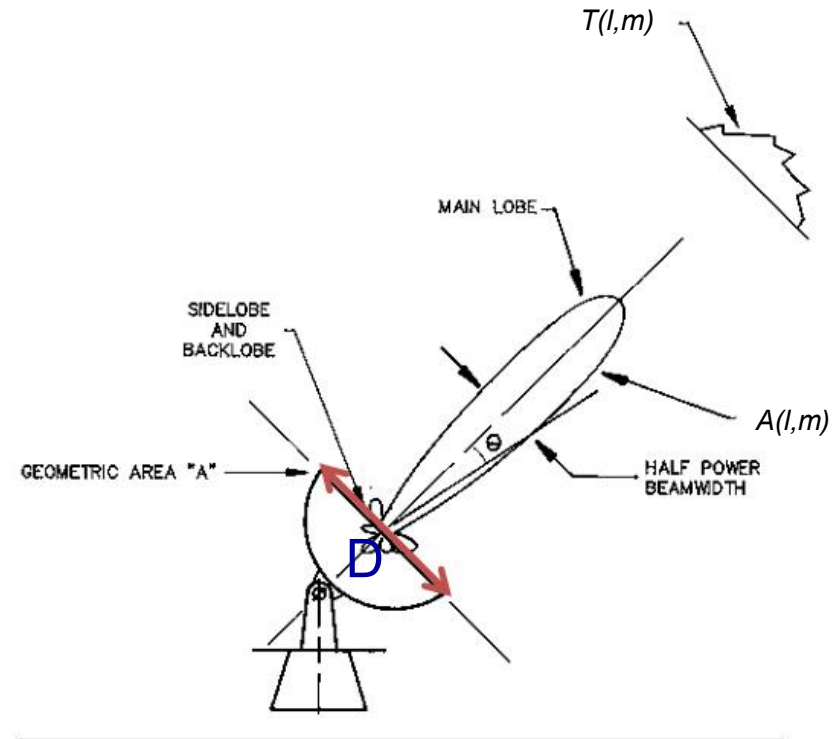
- “Fourier transform”
 - Fast Fourier Transform (FFT) algorithm is much faster than simple Fourier summation, $O(N \log N)$ for $2^N \times 2^N$ image
 - FFT requires data on a regularly spaced grid
 - aperture synthesis does not provide $V(u, v)$ on a regularly spaced grid, so...
- “gridding” used to resample $V(u, v)$ for FFT
 - customary to use a convolution method
 - special (“spheroidal”) functions that minimize smoothing and aliasing

$$V^G(u, v) = V(u, v)S(u, v) * G(u, v) \\ \xrightarrow{F} T^D(l, m)g(l, m)$$



Antenna Primary Beam Response

- antenna response $A(l,m)$ is not uniform across the entire sky
 - main lobe = “primary beam”
fwhm $\sim \lambda/D$
 - response beyond primary beam can be important (“sidelobes”)
- antenna beam modifies the sky brightness distribution
 - $T(l,m) \rightarrow T(l,m)A(l,m)$
 - can correct with division by $A(l,m)$ in the image plane
 - large source extents require multiple pointings of antennas = mosaicking



SMA 6 m
345 GHz

ALMA 12 m
690 GHz



Imaging Decisions: Pixel Size, Image Size

- pixel size
 - satisfy sampling theorem for longest baselines

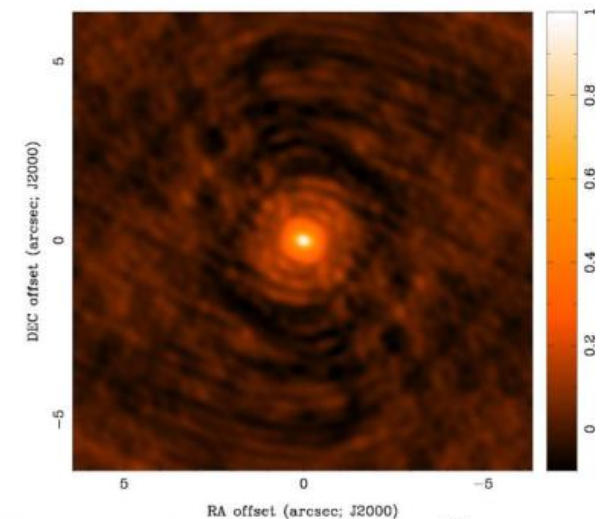
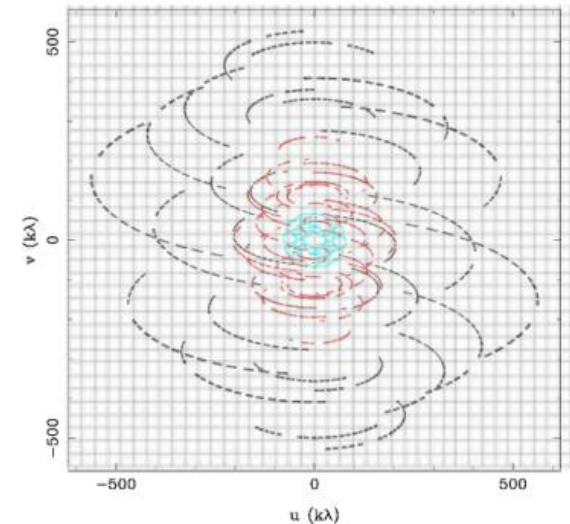
$$\Delta l < \frac{1}{2u_{max}} \quad \Delta m < \frac{1}{2v_{max}}$$

- in practice, 3 to 5 pixels across main lobe of dirty beam to aid deconvolution
 - e.g. at 870 μm with baselines to 500 meters $\rightarrow$ pixel size < 0.1 arcsec
 - CASA “cell” size
- image size
 - natural choice is often the full extent of the primary beam $A(l,m)$
 - e.g. SMA at 870 μm , 6 meter antennas $\rightarrow$ image size 2×35 arcsec
 - if there are bright sources in the sidelobes of $A(l,m)$, then the FFT will alias them into the image $\rightarrow$ make a larger image (or equivalent)
 - CASA “imsize”



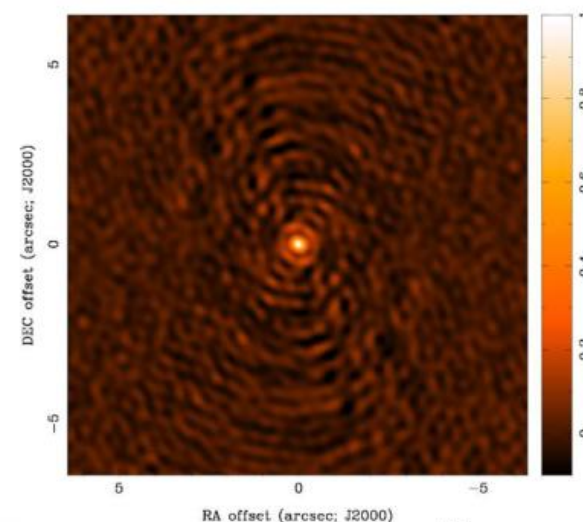
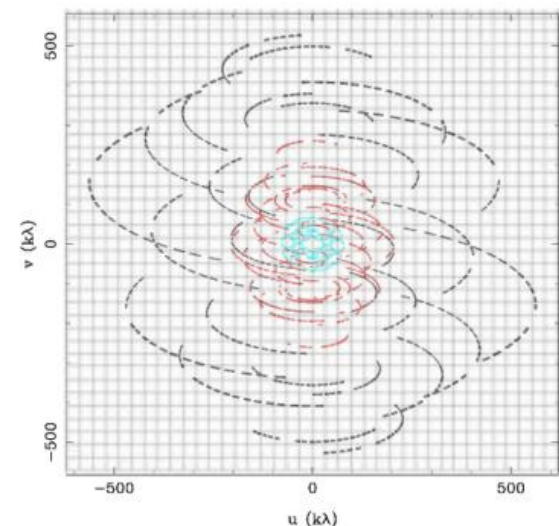
Imaging Decisions: Visibility Weighting

- introduce weighting function $W(u,v)$
 - modifies sampling function
 - $S(u,v) \rightarrow S(u,v)W(u,v)$
 - changes $s(l,m)$, the dirty beam shape
- natural weight
 - $W(u,v) = 1/\sigma^2$ in occupied (u,v) cells, where σ^2 is the noise variance, and $W(u,v) = 0$ everywhere else
 - maximizes point source sensitivity
 - lowest rms in image
 - generally gives more weight to short baselines (low spatial frequencies), so angular resolution is degraded



Dirty Beam Shape and Weighting

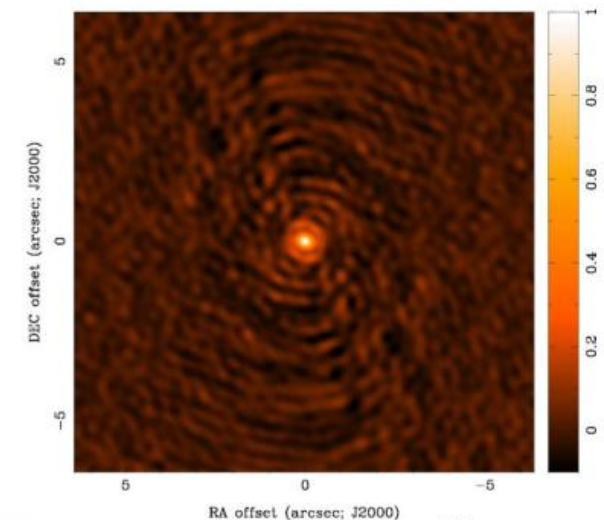
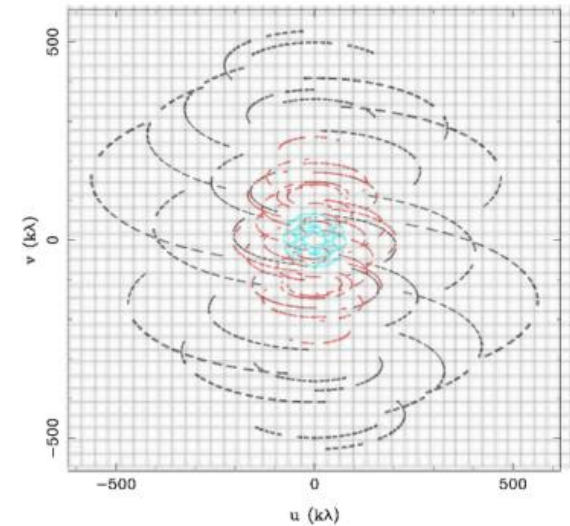
- uniform weight
 - $W(u,v)$ is inversely proportional to local density of (u,v) points
 - sum of weights in a (u,v) cell = const (and 0 for empty cells)
 - fills (u,v) plane more uniformly and dirty beam sidelobes are lower
 - gives more weight to long baselines (high spatial frequencies), so angular resolution is enhanced
 - downweights some data, so point source sensitivity is degraded
 - can be trouble with sparse sampling: cells with few data points have same weight as cells with many data points



Dirty Beam Shape and Weighting

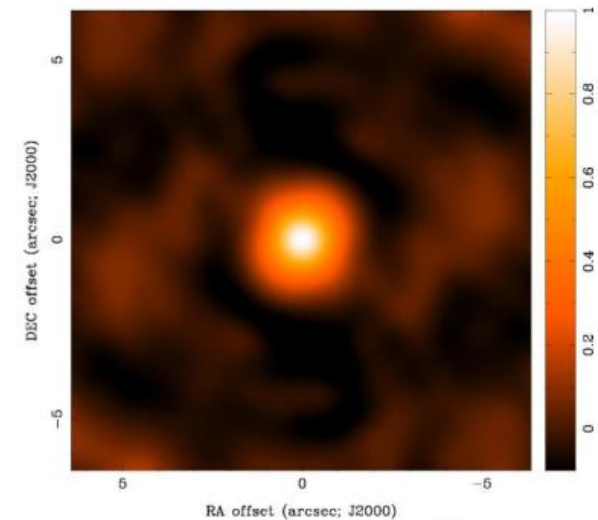
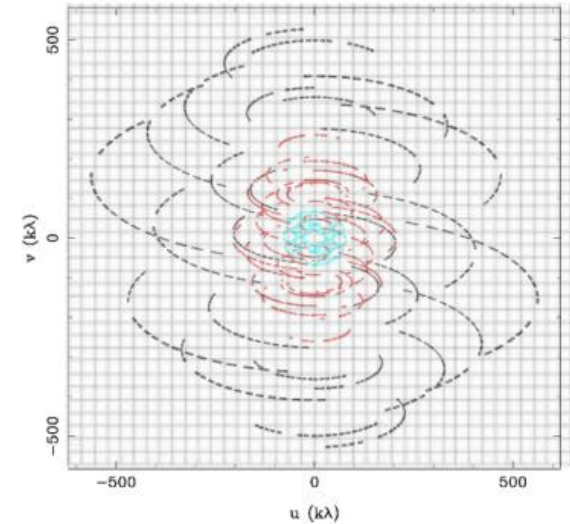
- robust (Briggs) weight
 - variant of uniform that avoids giving too much weight to (u,v) cells with low natural weight
 - software implementations differ
 - e.g.
$$W(u, v) = \frac{1}{\sqrt{1 + S_N^2/S_{thresh}^2}}$$

S_N is natural weight of cell
 S_{thresh} is a threshold
 high threshold $\rightarrow$ natural weight
 low threshold $\rightarrow$ uniform weight
- *an adjustable parameter allows for continuous variation between maximum point source sensitivity and resolution*



Dirty Beam Shape and Weighting

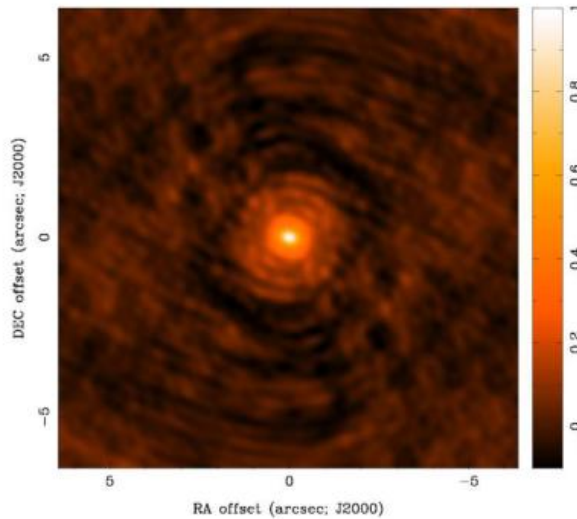
- tapering
 - apodize (u, v) sampling by a Gaussian
- $$W(u, v) = \exp\left(-\frac{(u^2 + v^2)}{t^2}\right)$$
- t = adjustable tapering parameter
- like smoothing in the image plane (convolution by a Gaussian)
 - gives more weight to short baselines, degrades angular resolution
 - downweights some data, so point source sensitivity degraded
 - may improve sensitivity to extended structure sampled by short baselines
 - limits to usefulness



Weighting and Tapering: Image Noise

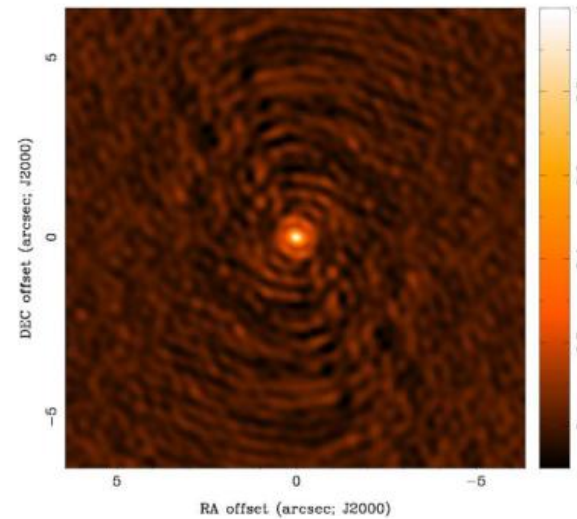
natural
0.59x0.50

rms=1.0



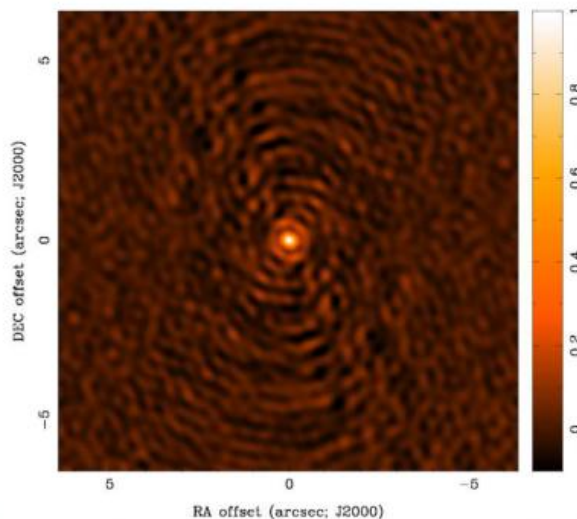
robust=0
0.40x0.34

rms=1.3



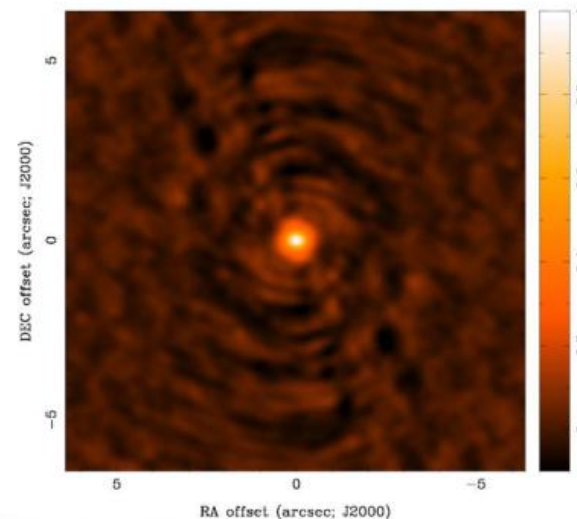
uniform
0.35x0.30

rms=2.1



robust=0
+ taper to
0.59x0.50

rms=1.2



Weighting and Tapering: Summary

- imaging parameters provide a lot of freedom
- appropriate choices depend on science goals

	Robust/Uniform	Natural	Taper
resolution	higher	medium	lower
sidelobes	lower	higher	depends
point source sensitivity	lower	maximum	lower
extended source sensitivity	lower	medium	higher



Beyond the Dirty Image: Deconvolution

- to keep you awake at night
 - $\exists$ an infinite number of $T(l,m)$ compatible with sampled $V(u,v)$, with “invisible” distributions $R(l,m)$ where $s(l,m) * R(l,m) = 0$
 - no data beyond $u_{\max}, v_{\max} \rightarrow$ unresolved structure
 - no data within $u_{\min}, v_{\min} \rightarrow$ limit on largest size scale
 - holes in between $\rightarrow$ synthesized beam sidelobes
 - noise $\rightarrow$ undetected/corrupted structure in $T(l,m)$
 - no unique prescription for extracting optimum estimate of $T(l,m)$
- deconvolution
 - uses non-linear techniques to interpolate/extrapolate samples of $V(u,v)$ into unsampled regions of the (u,v) plane
 - aims to find a sensible model of $T(l,m)$ compatible with data
 - requires *a priori* assumptions about $T(l,m)$ to pick plausible “invisible” distributions to fill unmeasured parts of the Fourier plane



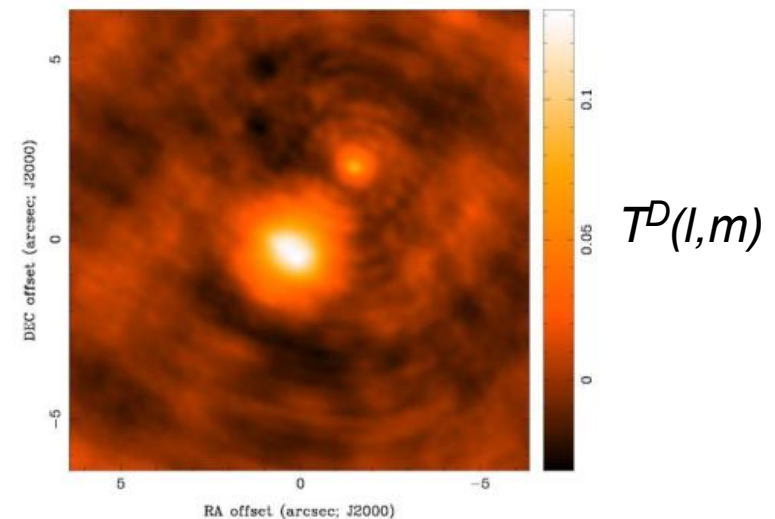
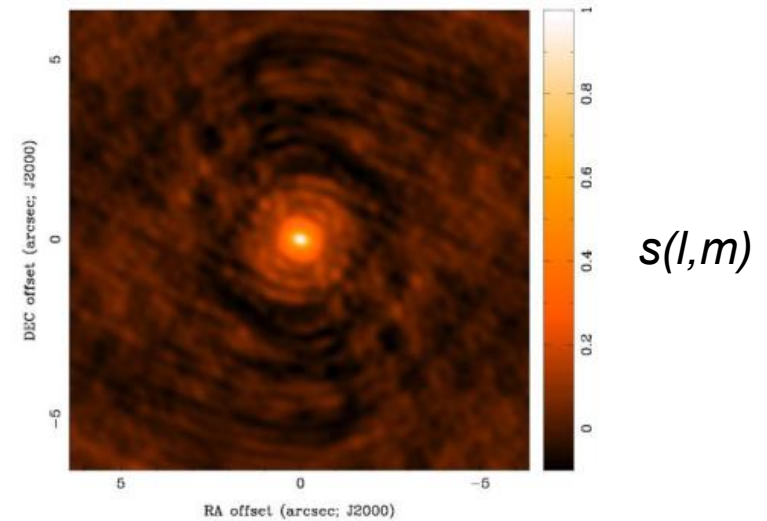
Deconvolution Algorithms

- an active research area, e.g. compressive sensing methods
- **clean**: dominant deconvolution algorithm in radio astronomy
 - *a priori* assumption: $T(l,m)$ is a collection of point sources
 - fit and subtract the synthesized beam iteratively
 - original version by Högbom (1974) purely image based
 - variants developed for higher computational efficiency, model visibility subtraction, to deal better with extended emission structure, etc.
- **maximum entropy**: a rarely used alternative
 - *a priori* assumption: $T(l,m)$ is smooth and positive
 - define “smoothness” via a mathematical expression for entropy, e.g. Gull and Skilling (1983), find smoothest image consistent with data
 - vast literature about the deep meaning of entropy as information content



Basic clean Algorithm

- initialize
 - a *residual map* to the dirty map
 - a *Clean Component* list
- 1. identify the highest peak in the residual map as a point source
- 2. subtract a fraction of this peak from the *residual map* using a scaled dirty beam, $s(l,m)$ x gain
- 3. add this point source location and amplitude to the *Clean Component* list
- 4. goto step 1 (an iteration) unless stopping criterion reached



Basic clean Algorithm (continued)

- stopping criteria?
 - *residual map* maximum < threshold = multiple of rms (if noise limited)
 - *residual map* maximum < threshold = fraction of dirty map maximum (if dynamic range limited)
 - maximum number of *Clean Components* reached (no justification)
- loop gain?
 - good results for $g=0.1$ to 0.3
 - lower values can work better for smoother emission, $g=0.05$
- easy to include *a priori* information about where in dirty map to search for *Clean Components* (using “boxes” or “masks”)
 - very useful but potentially dangerous
- Schwarz (1978) showed that the clean algorithm is equivalent to a least squares fit of sinusoids to visibilities in the case of no noise

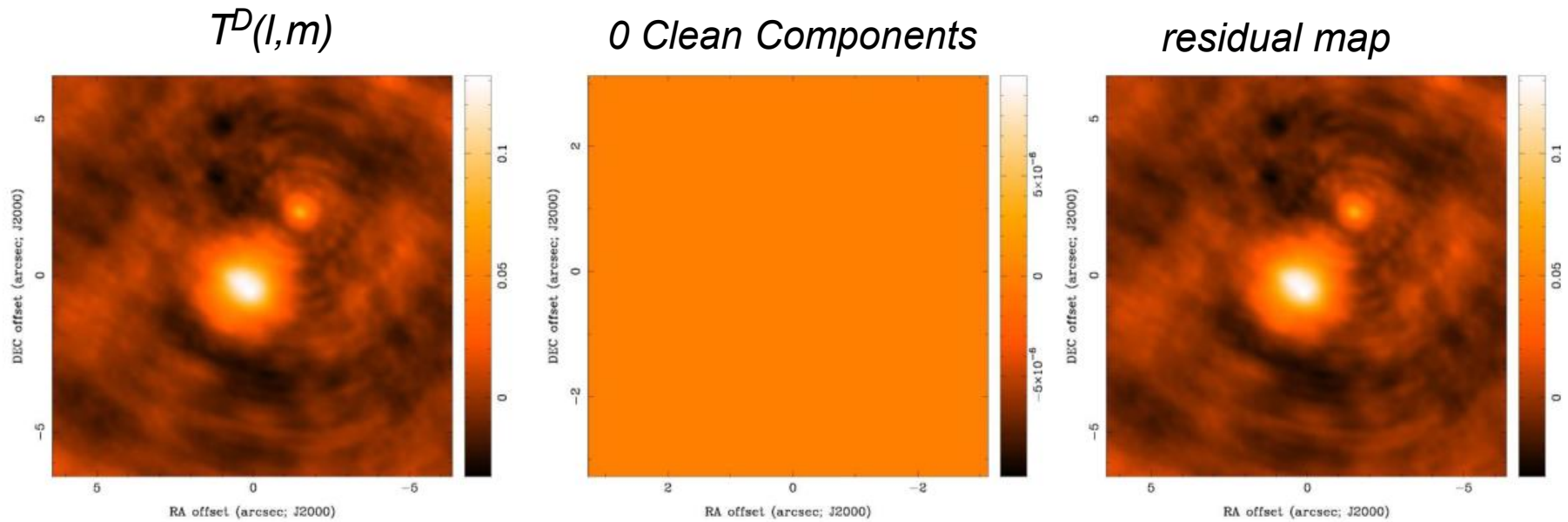


Basic clean Algorithm (continued)

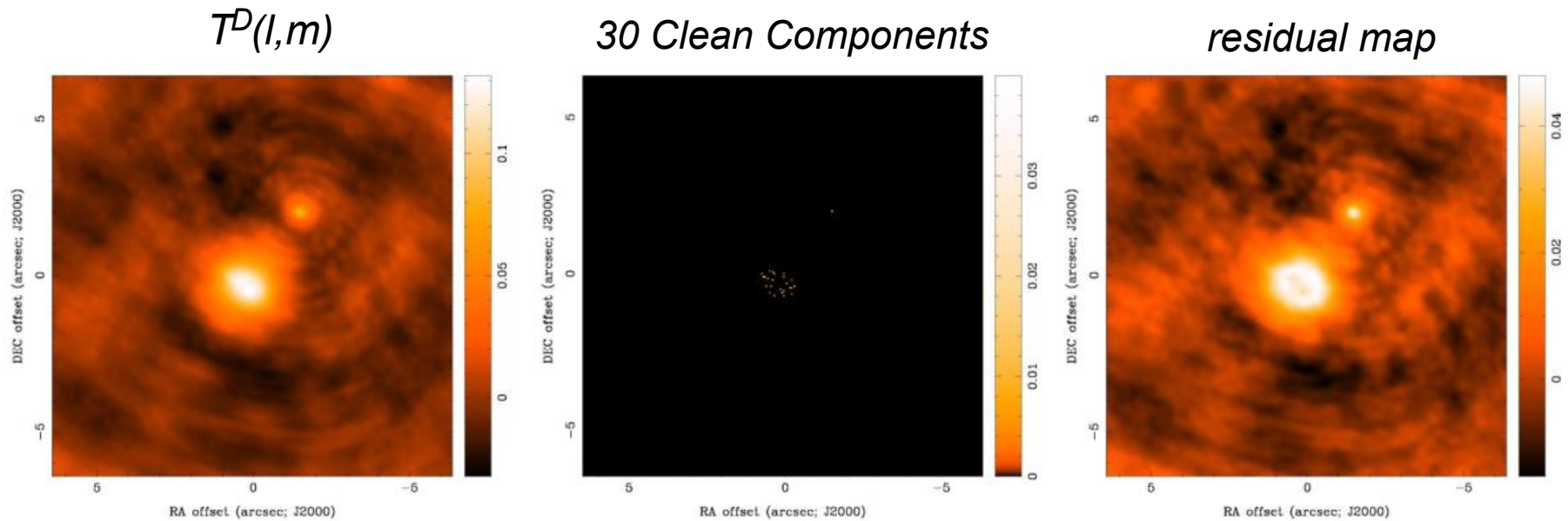
- last step: make “restored” image
 - make a model image with all point source *Clean Components*
 - convolve point sources with an elliptical Gaussian, fit to the main lobe of the dirty beam (“clean beam”); avoids super-resolution of model
 - add *residual map* of noise and source structure below the threshold
- resulting restored image is an estimate of the true sky brightness $T(l,m)$
- units of the restored image are (mostly) Jy per clean beam area
= intensity (or brightness temperature)
- for most weighting schemes, there is information in the image from baselines that sample high spatial frequencies within the clean beam fwhm, so modest super-resolution may be OK
- the restored image does not actually fit the observed visibilities



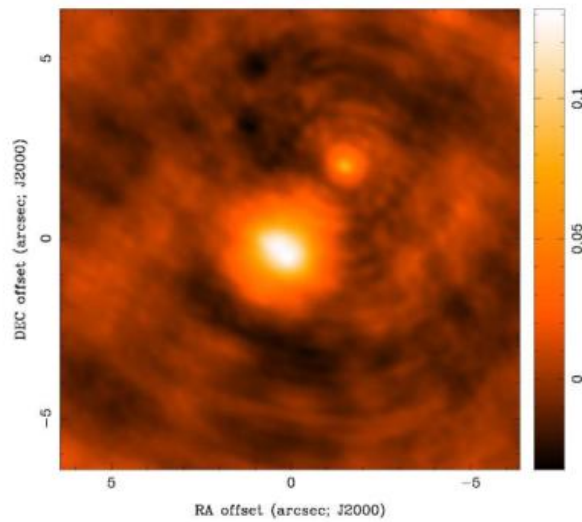
clean Example



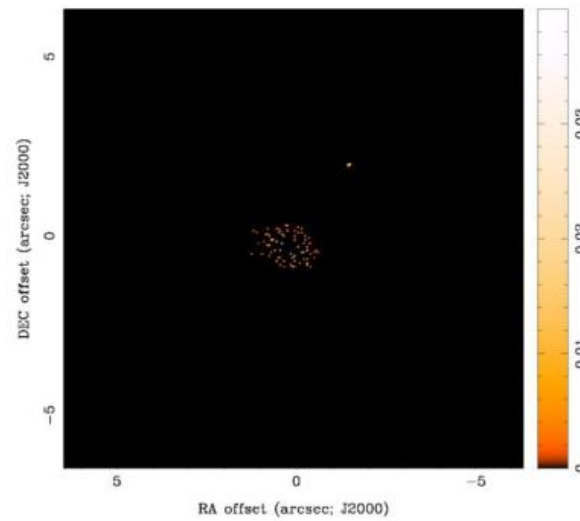
clean Example



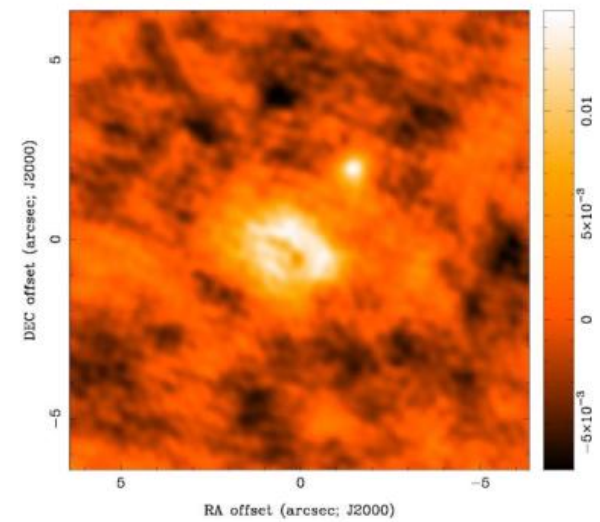
clean Example

 $T^D(l, m)$ 

100 Clean Components

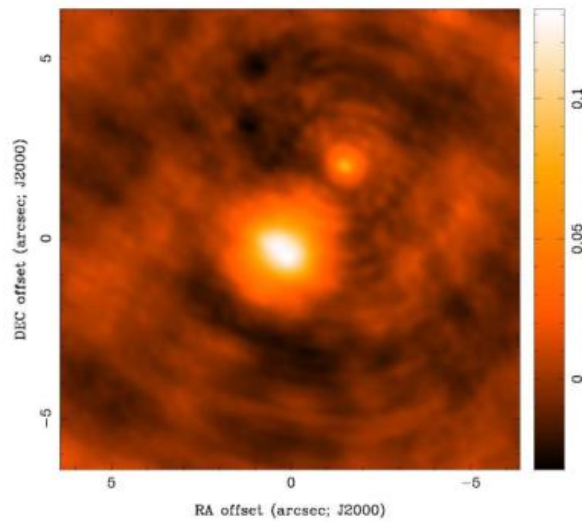


residual map

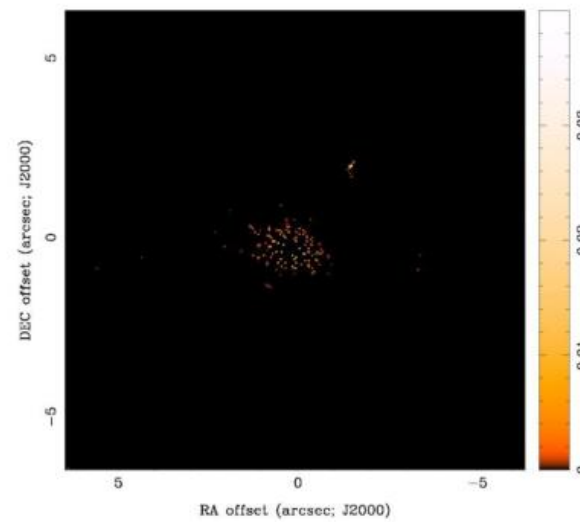


clean Example

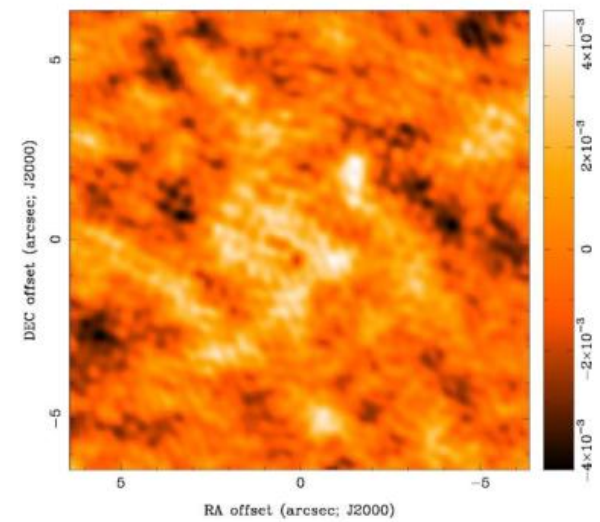
$T^D(l,m)$



300 Clean Components

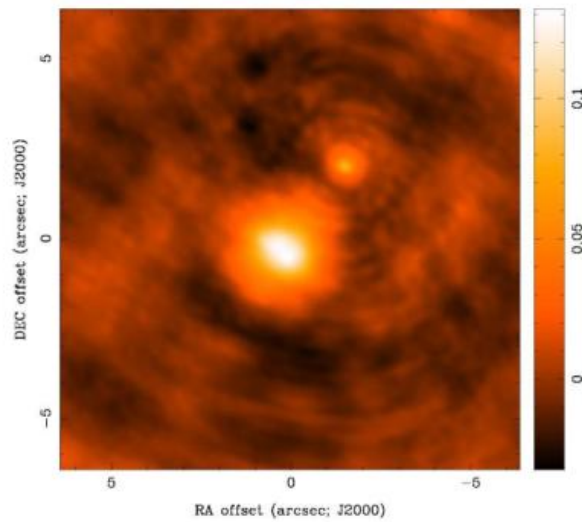


residual map

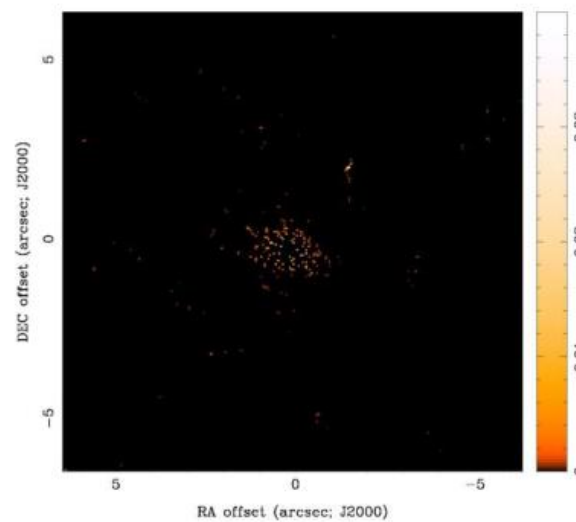


clean Example

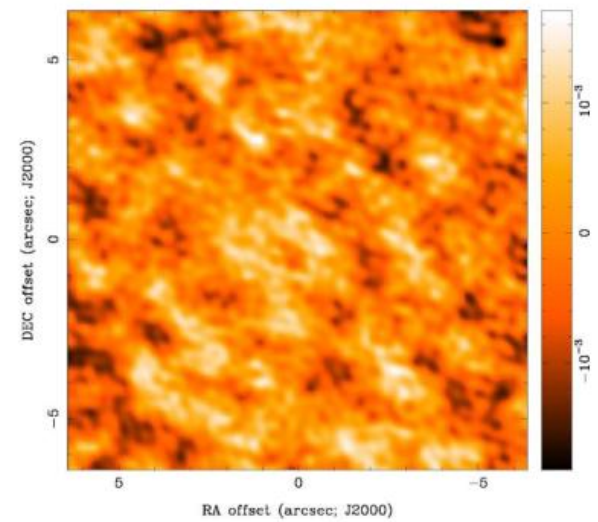
$T^D(l,m)$



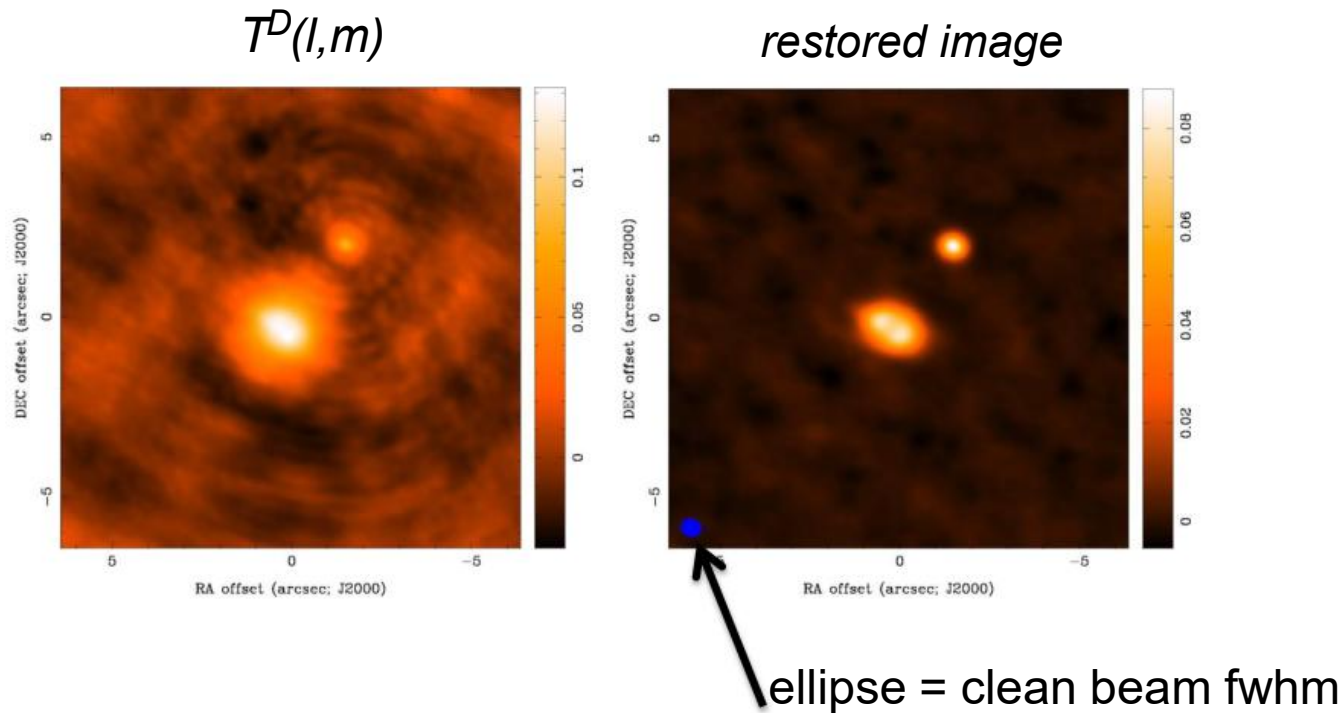
583 Clean Components



residual map



clean Example



final image depends on

*imaging parameters (pixel size, visibility weighting scheme, gridding)
and deconvolution (algorithm, iterations, masks, stopping criteria)*



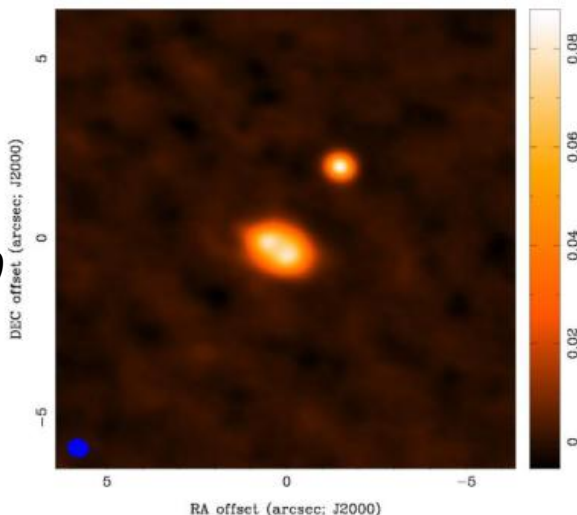
CASA clean filename extensions

- `<imagename>.image`
 - final clean image (or dirty image if `niter=0`)
- `<imagename>.psf`
 - point spread function (= dirty beam)
- `<imagename>.model`
 - image of clean components
- `<imagename>.residual`
 - residual after subtracting clean components
(use to decide whether or not to continue clean)
- `<imagename>.flux`
 - relative sensitivity on the sky
 - `pbcor = True` divides `.image` by `.flux`

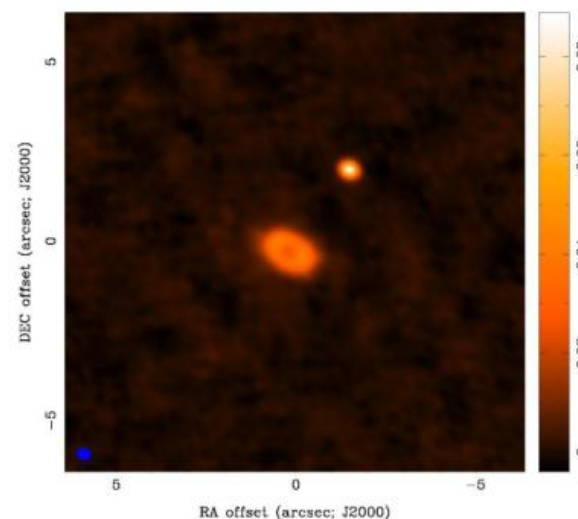


Results from Different Weighting Schemes

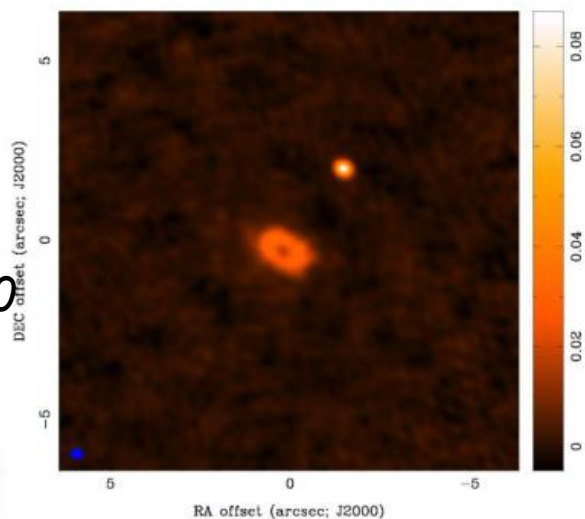
natural
 0.59×0.50



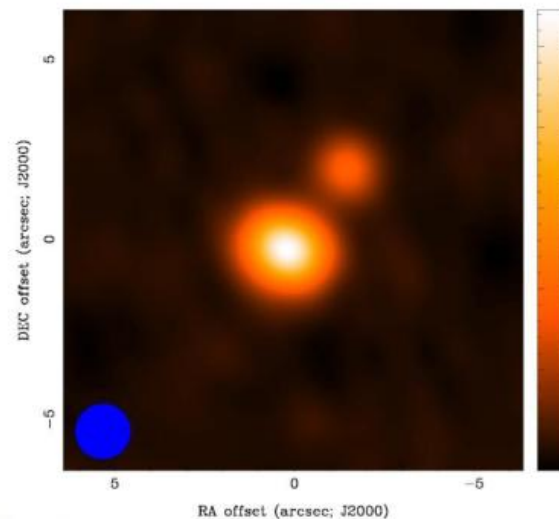
robust=0
 0.40×0.34



uniform
 0.35×0.30

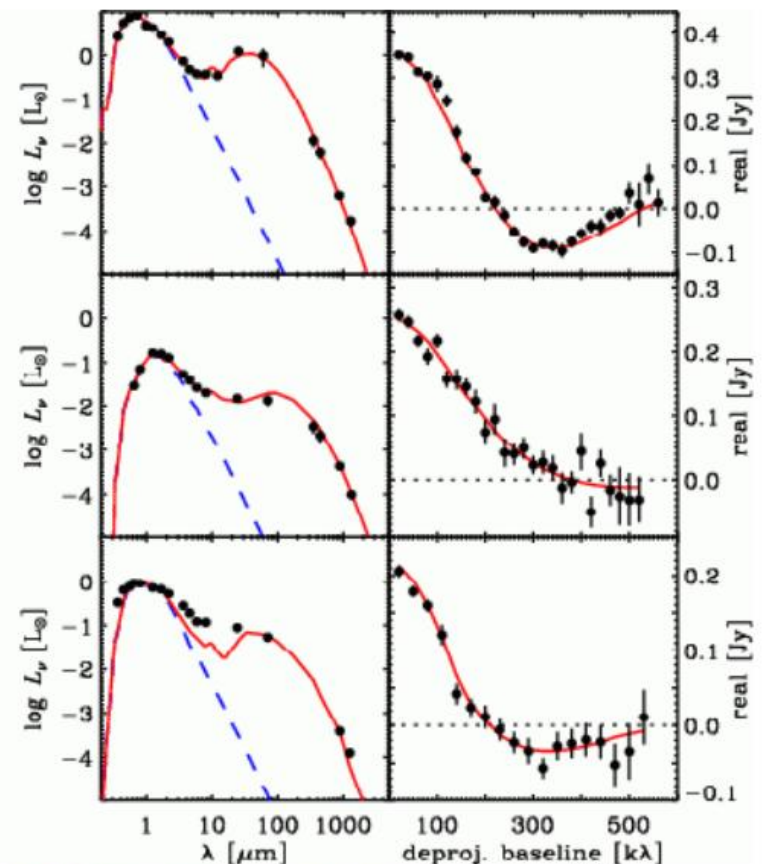
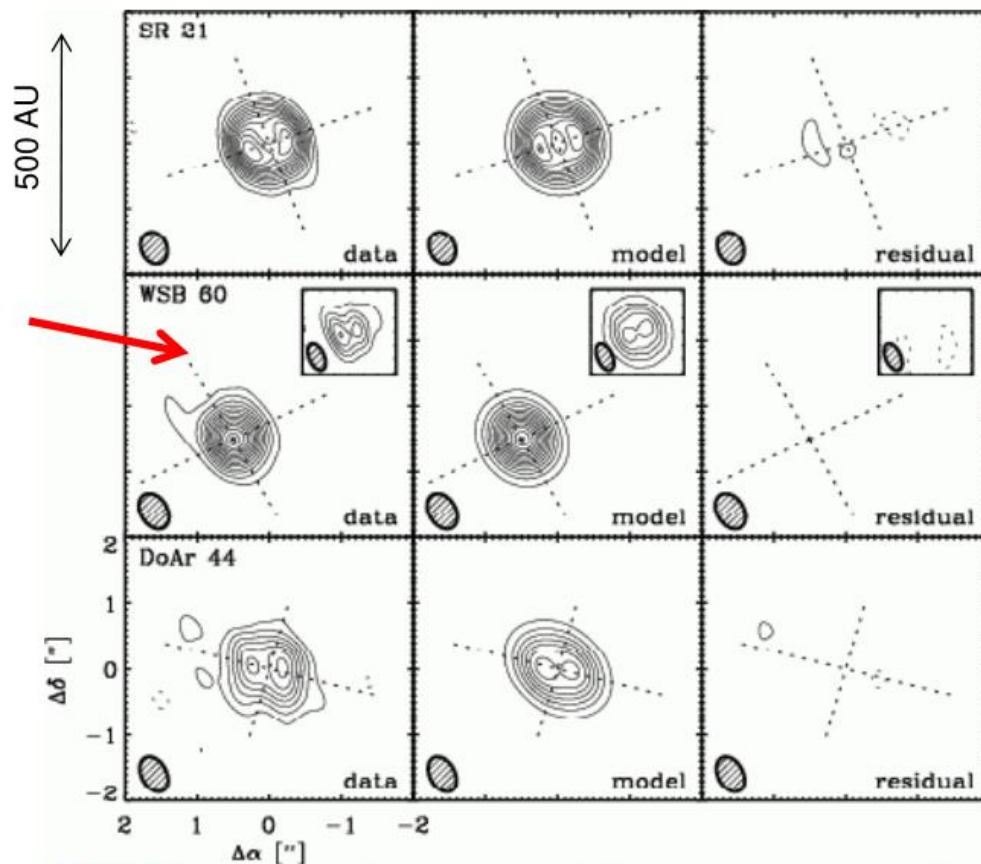


natural
+ taper to
 1.5×1.5



Tune Resolution/Sensitivity to suit Science

- example: SMA 870 μm images of protoplanetary disk dust continuum emission with resolved inner cavities (Andrews et al. 2009, ApJ, 700, 1502)



Scale Sensitive Deconvolution Algorithms

- basic clean (or Maximum Entropy) is scale-free and treats each pixel as an independent degree of freedom: no concept of source size
- adjacent pixels in an image are not independent
- an extended source covering 1000 pixels might be characterized by just a few parameters, not 1000 parameters (e.g. an elliptical Gaussian with 6 parameters: x, y, amp, major fwhm, minor fwhm, position angle)
- scale sensitive deconvolution algorithms try to employ fewer degrees of freedom to model plausible sky brightness distributions
- MS Clean (Multi-Scale Clean)
- Adaptive Scale Pixel (Asp) Clean
- yields promising results on extended emission



“Invisible” Large Scale Structure

- missing short spacings can be problematic for large scale structure
- to estimate? simulate observations, or check simple expressions for a Gaussian or uniform disk (appendix of Wilner & Welch 1994, ApJ, 427, 898)

Homework Problem

- Q: By what factor is the central brightness reduced as a function of source size due to missing short spacings for a Gaussian characterized by fwhm $\theta_{1/2}$?
- A: a Gaussian source central brightness is reduced 50% when

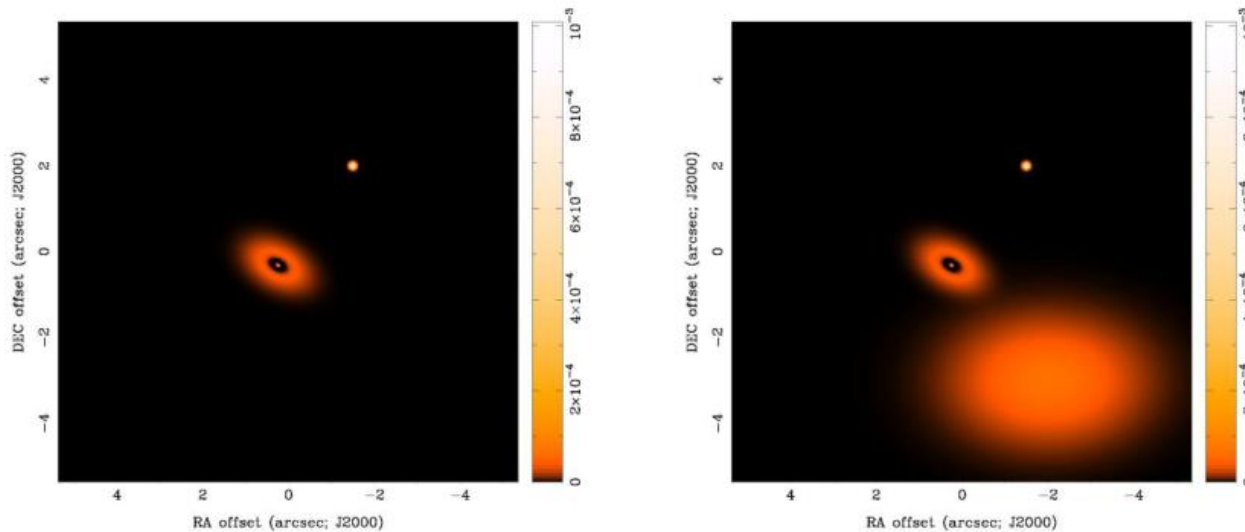
$$\theta_{1/2} = 18'' \left(\frac{\nu}{100 \text{ GHz}} \right)^{-1} \left(\frac{B_{min}}{15 \text{ meters}} \right)^{-1}$$

where B_{min} is the shortest baseline [meters], ν is the frequency [GHz]



Missing Short Spacings: Demonstration

- important structure may be missed in central hole of (u,v) coverage
- Do the visibilities observed in our example discriminate between these two models of the sky brightness distribution $T(l,m)$?



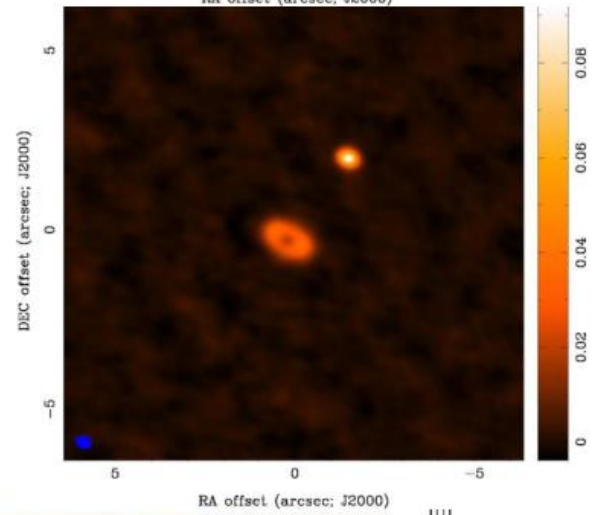
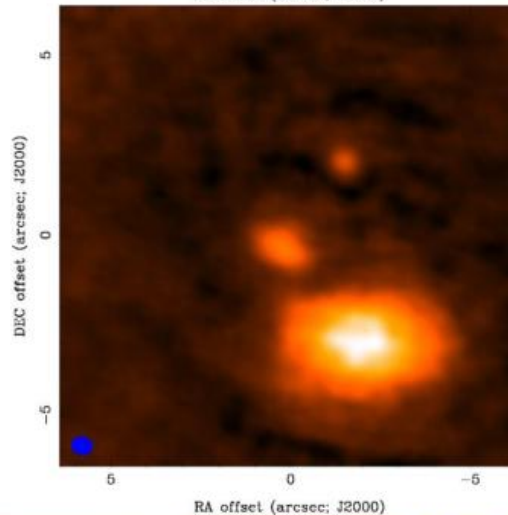
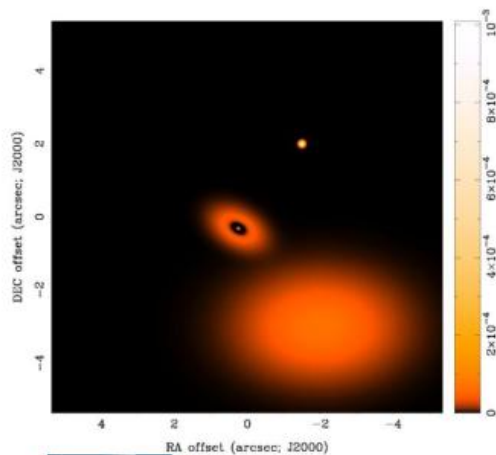
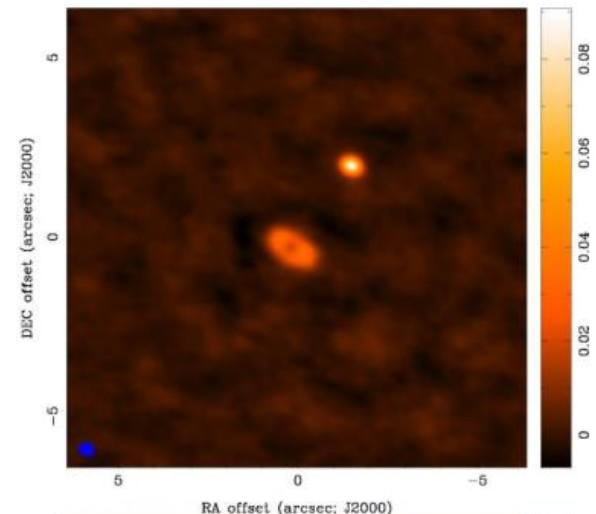
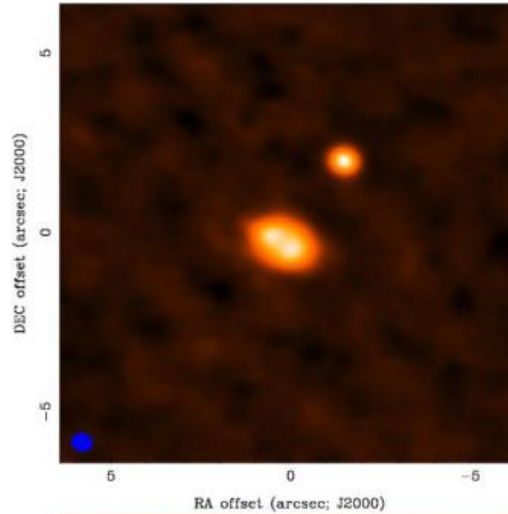
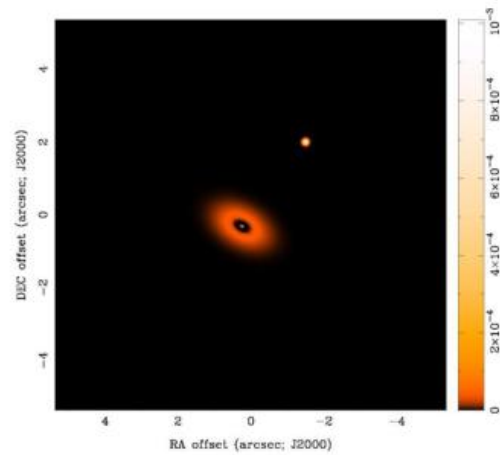
- Yes... but only on baselines shorter than about $75 \text{ k} \lambda$

Missing Short Spacings: Demonstration

$T(l,m)$

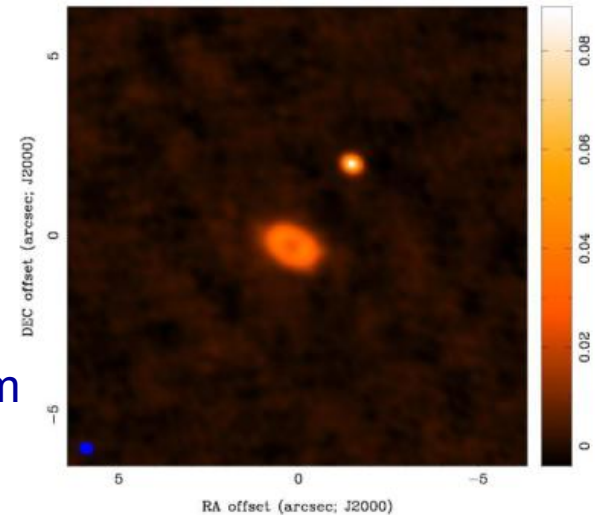
natural weight

$> 75\text{ k}\lambda$ *natural weight*



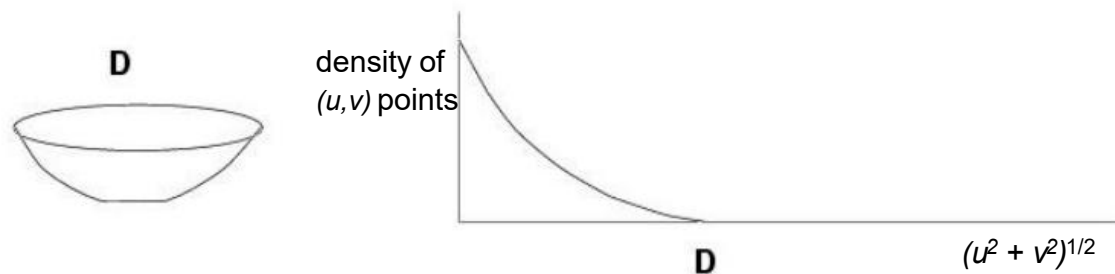
Measures of Image Quality

- *dynamic range*
 - ratio of peak brightness to rms noise in a region void of emission
 - easy way to calculate a lower limit to the error in brightness in a non-empty region
 - e.g. peak = 89 mJy/beam, rms = 0.9 mJy/beam
 $\rightarrow DR = 89/0.9 = 99$
- *fidelity*
 - difference between any produced image and the correct image
 - fidelity image = input model / difference
 - = $\text{model} * \text{beam} / \text{abs}(\text{model} * \text{beam} - \text{reconstruction})$
 - = inverse of the relative error
 - need knowledge of the correct image to calculate



Techniques to Obtain Short Spacings

use a large single dish telescope



- all Fourier components from 0 to D sampled, where D is dish diameter (weighting depends on illumination)
- scan single dish across sky to make an image $T(l,m) * A(l,m)$
where $A(l,m)$ is the single dish response pattern
- Fourier transform single dish image, $T(l,m) * A(l,m)$, to get $V(u,v)a(u,v)$
and then divide by $a(u,v)$ to estimate $V(u,v)$ for baselines $< D$
- choose D large enough to overlap interferometer samples of $V(u,v)$
and avoid using data where $a(u,v)$ becomes small, e.g. VLA & GBT



Techniques to Obtain Short Spacings

use a separate array of smaller antennas

- small antennas can observe short baselines inaccessible to larger ones
- the larger antennas can be used as single dish telescopes to make images with Fourier components not accessible to the smaller antennas
- example: ALMA main array + ACA

main array

50 x 12m: 12m to 14+ km

ACA

12 x 7m: covers 7-12m

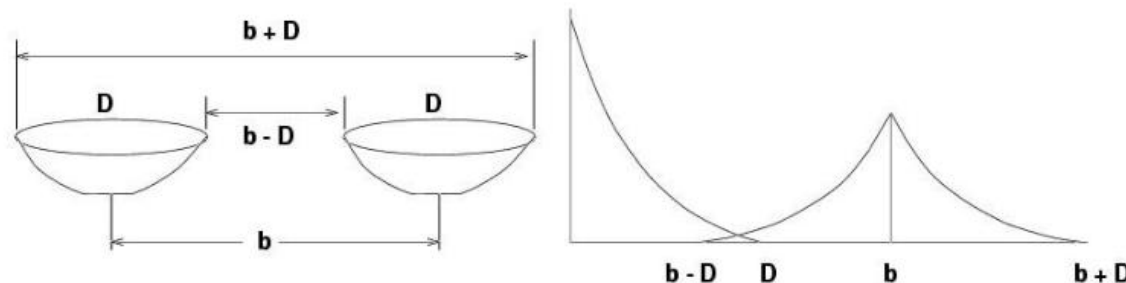
4 x 12m single dishes: 0-7m



Techniques to Obtain Short Spacings

mosaic with a homogeneous array

- recover a range of spatial frequencies around the nominal baseline b using knowledge of $A(l,m)$, shortest spacings from single dishes (Ekers & Rots 1979)



- $V(u,v)$ is a linear combination of baselines from $b-D$ to $b+D$
- depends on pointing direction (l_0, m_0) as well as on (u,v)

$$V(u, v; l_0, m_0) = \int \int T(l, m) A(l - l_0, m - m_0) e^{i2\pi(ul + vm)} dl dm$$

- Fourier transform with respect to pointing direction (l_0, m_0)

$$V(u - u_0, v - v_0) = \left(\int \int V(u, v; l_0, m_0) e^{i2\pi(u_0 l_0 + v_0 m_0)} dl_0 dm_0 \right) / a(u_0, v_0)$$



Self Calibration

- *a priori* calibration using external calibrators is not perfect
 - interpolated from different time, different sky direction from source
- basic idea of self calibration is to correct for antenna based phase and amplitude errors *together* with imaging to create a source model
- works because
 - at each time, measure N complex gains and $N(N-1)/2$ visibilities
 - source structure can be represented by a small number of parameters
 - a highly overconstrained problem if N large and source simple
- in practice, an iterative, non-linear relaxation process
 - assume source model $\rightarrow$ solve for time dependent gains $\rightarrow$ form new source model from corrected data using e.g. clean $\rightarrow$ solve for new gains
 - requires sufficient signal-to-noise at each solution interval
- loses absolute phase from calibrators and therefore position information
- dangerous with small N arrays, complex sources, marginal signal-to-noise



Concluding Remarks

- interferometry samples Fourier components of sky brightness
- make an image by Fourier transforming sampled visibilities
- deconvolution attempts to correct for incomplete sampling
- remember
 - there are an infinite number of images compatible with the visibilities
 - missing (or corrupted) visibilities affect the entire image
- astronomers must use judgement in the imaging and deconvolution process
- it's fun and worth the trouble → high angular resolution images!

many, many issues not covered in this talk: see References and upcoming talks



END

